

NCERT Exemplar Solutions for Class 12

Mathematics

Chapter 9 – Differential Equations

Solved Examples

Short Answer Questions

1. Find the differential equation of the family of curves $y = Ae^{2x} + Be^{-2x}$

Ans: We have given that $y = Ae^{2x} + Be^{-2x}$

Now, differentiate both sides with respect to x we get:

$$\Rightarrow \frac{dy}{dx} = 2Ae^{2x} - 2Be^{-2x}$$

$$\Rightarrow \frac{d^2y}{dx^2} = 4Ae^{2x} + 4Be^{-2x}$$

$$\Rightarrow \frac{d^2y}{dx^2} = 4y$$

$$\Rightarrow \frac{d^2y}{dx^2} - 4y = 0$$

2. Find the general solution of the differential equation $\frac{dy}{dx} = \frac{y}{x}$

Ans: We have given that $\frac{dy}{dx} = \frac{y}{x}$

$$\text{implies } \frac{dy}{y} = \frac{dx}{x}$$

$$\int \frac{dy}{y} = \int \frac{dx}{x}$$

Now, use formula i.e. $\log A + \log B = \log AB$ in RHS and then remove log, we get:

$$\Rightarrow \log y = \log x + \log c$$

$$\Rightarrow y = cx$$

3. Given that $\frac{dy}{dx} = ye^x$ and $x=0, y=e$. Find the value of y when $x=1$

Ans: We have given that $\frac{dy}{dx} = ye^x$ which implies

$$\int \frac{dy}{y} = \int e^x dx$$

$$\Rightarrow \log y = e^x + c$$

Now, substitute $x=0, y=e$ we get:

$$\log e = e^0 + c$$

$$\text{i.e. } c = 0 \text{ as } \log e = 1, e^0 = 1$$

$$\text{So, } \log y = e^x$$

Put $x=1$ we get

$$\Rightarrow \log y = e$$

$$\Rightarrow y = e^e$$

4. Solve the differential equation $\frac{dy}{dx} + \frac{y}{x} = x^2$

Ans: This equation is of type $\frac{dy}{dx} + Py = Q$ which is a differential equation

$$\text{I.F.} = \int \frac{1}{x} dx = e^{\log x} = x \times (e^{\log(x)} = f(x))$$

$$\text{So, solution of given differential equation is } yx = \int xx^2 dx \Rightarrow yx = \frac{x^4}{4} + c$$

Thus, $y = \frac{x^3}{4} + \frac{c}{x}$

5. Find the differential equation of the family of lines through the origin.

Ans: Let $y = mx$ represents family of lines through origin i.e. $\frac{dy}{dx} = m$

Now, Eliminate m we get:

$$\Rightarrow y = \frac{dy}{dx} \times x$$

$$\Rightarrow x \frac{dy}{dx} - y = 0$$

6. Find the differential equation of all non-horizontal lines in a plane.

Ans: Here, general equation of all non-horizontal lines in a plane represents as $ax + by = c, a \neq 0$

Now differentiate both sides with respect to y we get:

$$\Rightarrow a \frac{dx}{dy} + b = 0$$

Again, differentiate with respect to y we get:

$$\Rightarrow \frac{d^2x}{dy^2} = 0$$

$$\Rightarrow \frac{dx^2}{dy^2} = 0$$

7. Find the equation of a curve whose tangent at any point on it, different from origin has slope $y + \frac{y}{x}$

Ans: We have given that $\frac{dy}{dx} = y + \frac{y}{x} = y\left(1 + \frac{1}{x}\right)$

$$\text{implies } \frac{dy}{y} = \left(1 + \frac{1}{x}\right) dx$$

Now, integrate both sides, we get:

$$\Rightarrow \int \frac{dy}{y} = \int \left(1 + \frac{1}{x}\right) dx$$

$$\Rightarrow \log y = x + \log x + c$$

$$\Rightarrow \log\left(\frac{y}{x}\right) = x + c$$

$$\Rightarrow \frac{y}{x} = e^{x+c} = e^x \times e^c$$

$$\Rightarrow \frac{y}{x} = k \times e^x$$

$$\Rightarrow y = kx \times e^x$$

Long Answer Questions

8. Find the equation of a curve which is passing through the point (1,1) .If the perpendicular distance of the origin from the normal at any point P(x,y) of the curve is equal to the distance of P from the x -axis.

Ans: Here, let equation of normal at P(x,y) be $\Rightarrow Y - y = \frac{-dx}{dy}(X-x)$

$$Y + X \frac{dx}{dy} - \left(y + x \frac{dx}{dy}\right) = 0 \dots\dots\dots(i)$$

So, length of perpendicular from origin to (i) be $\frac{y+x \frac{dx}{dy}}{\sqrt{1+\left(\frac{dx}{dy}\right)^2}} \dots\dots\dots(ii)$

Also, distance between P and x -axis is |y|

Then, we have

$$\frac{y+x \frac{dx}{dy}}{\sqrt{1+\left(\frac{dx}{dy}\right)^2}} = |y|$$

$$\Rightarrow \left(y+x \frac{dx}{dy}\right)^2 = y^2 \left[1+\left(\frac{dx}{dy}\right)^2\right]$$

$$\Rightarrow \frac{dx}{dy} \left[\frac{dx}{dy} (x^2 - y^2) + 2xy \right] = 0$$

$$\Rightarrow \frac{dx}{dy} = 0$$

$$\Rightarrow \frac{dx}{dy} = \frac{2xy}{y^2 - x^2}$$

Case I: $\frac{dx}{dy} = 0 \Rightarrow dx = 0$

Now, integrate both sides we have $x = k$

Substitute $x = 1$ we get $k = 1$

So, $x = 1$ is equation of curve which is not possible. Thus, rejected

Case II: $\frac{dx}{dy} = \frac{2xy}{y^2 - x^2} \Rightarrow \frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$

Substitute $y = vx$ we have

$$v + x \frac{dv}{dx} = \frac{v^2 x^2 - x^2}{2vx^2}$$

$$\Rightarrow x \times \frac{dv}{dx} = \frac{v^2 - 1}{2v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v^2 - 1}{2v} - v$$

$$= \frac{-(1+v^2)}{2v} \Rightarrow \frac{2v}{1+v^2} dv = \frac{-dx}{x}$$

Now, integrate both sides:

$$\log(1+v^2) = -\log x + \log c$$

$$\Rightarrow \log(1+v^2)(x) = \log c(1+v^2)x = c$$

$$\Rightarrow x^2 + y^2 = cx$$

Substitute $x = 1, y = 1$ we have $c = 2$

$$\text{Thus, } x^2 + y^2 - 2x = 0$$

9. Find the equation of a curve passing through $\left(1, \frac{\pi}{4}\right)$. If the slope of the tangent to the curve at any point $P(x, y)$ is $\frac{y}{x} - \cos^2 \frac{y}{x}$

Ans: Here, slope of tangent $m = \frac{dy}{dx}$

$$\frac{dy}{dx} = \frac{y}{x} - \cos^2 \frac{y}{x} \quad (i)$$

which is a homogeneous equation

Now, substitute $y = vx$ we get:

$$v + x \frac{dv}{dx} = v - \cos^2 v$$

$$\Rightarrow x \frac{dv}{dx} = -\cos^2 v$$

$$\Rightarrow \sec^2 v \, dv = -\frac{dx}{x}$$

$$\Rightarrow \tan v = -\log x + c \quad \int \sec^2 v \, dv = \int -\frac{dx}{x}$$

$$\Rightarrow \tan \frac{y}{x} + \log x = c \quad (ii)$$

Again substitute $x = 1, y = \frac{\pi}{4}$ then we get $c = 1$

$\tan\left(\frac{y}{x}\right) + \log x = 1$ is the required equation.

10. Solve $x^2 \frac{dy}{dx} - xy = 1 + \cos\left(\frac{y}{x}\right)$, $x \neq 0$, $x = 1$, $y = \frac{\pi}{2}$

Ans: We have $x^2 \frac{dy}{dx} - xy = 2\cos^2\left(\frac{y}{2x}\right)$, $x \neq 0$

$$\Rightarrow \frac{x^2 \frac{dy}{dx} - xy}{2\cos^2\left(\frac{y}{2x}\right)} = 1$$

$$\Rightarrow \frac{\sec^2\left(\frac{y}{2x}\right)}{2} \left[x^2 \frac{dy}{dx} - xy \right] = 1$$

Now divide both sides by x^3 we get:

$$\Rightarrow \frac{\sec^2\left(\frac{y}{2x}\right)}{2} \left[\frac{x \frac{dy}{dx} - y}{x^2} \right] = \frac{1}{x^3}$$

$$\Rightarrow \frac{d}{dx} \left[\tan\left(\frac{y}{2x}\right) \right] = \frac{1}{x^3}$$

Now, integrate both sides

$$\tan\left(\frac{y}{2x}\right) = \frac{-1}{2x^2} + k$$

Substitute $x = 1$, $y = \frac{\pi}{2}$ we get $k = \frac{3}{2}$

$$\text{Thus, we have } \tan\left(\frac{y}{2x}\right) = \frac{1}{2x^2} + \frac{3}{2} \tan\left(\frac{y}{2x}\right) = \frac{-1}{2x^2} + \frac{3}{2}$$

11. State the type of the differential equation for the equation $x dy - y dx = \sqrt{x^2 + y^2} dx$ and solve it.

Ans: we have $x dy = (\sqrt{x^2 + y^2} + y) dx$

$$\text{i.e. } \frac{dy}{dx} = \frac{\sqrt{x^2 + y^2} + y}{x} \quad (i)$$

Thus, RHS is homogeneous of degree zero. Hence this is a homogeneous differential equation.

Now, substitute $y = vx$ we get

$$\begin{aligned} v+x \frac{dv}{dx} &= \frac{\sqrt{x^2 + v^2 x^2} + vx}{x} \\ \Rightarrow v+x \frac{dv}{dx} &= \sqrt{1+v^2} + v \\ \Rightarrow \frac{dv}{dx} &= \sqrt{1+v^2} \\ \Rightarrow \frac{dv}{\sqrt{1+v^2}} &= \frac{dx}{x} \quad (ii) \end{aligned}$$

Now integrate both sides

$$\begin{aligned} \log(v + \sqrt{1+v^2}) &= \log x + \log c \\ \Rightarrow v + \sqrt{1+v^2} &= cx \\ \Rightarrow \frac{y}{x} + \sqrt{1 + \frac{y^2}{x^2}} &= cx \\ \Rightarrow y + \sqrt{x^2 + y^2} &= cx^2 \end{aligned}$$

Objective Questions

Choose the correct answer form the given four options in each of the Examples 12 to 21

12. The degree of the differential equation $\left(1 + \frac{dy}{dx}\right)^3 = \left(\frac{d^2y}{dx^2}\right)^2$ is

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Ans: Correct option (B)

13. The degree of the differential equation $\frac{d^2y}{dx^2} + 3\left(\frac{dy}{dx}\right)^2 = x^2 \log\left(\frac{d^2y}{dx^2}\right)$ is

- (A) 1
- (B) 2
- (C) 3
- (D) Not defined

Ans: Correct option (D) because given differential equation is not a polynomial equation in terms of its derivatives, so its degree is not defined.

14. The order and degree of the differential equation $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^2 = \frac{d^2y}{dx^2}$ respectively, are

- (A) 1,2
- (B) 2,2
- (C) 2,1
- (D) 4,2

Ans: Correct option (C)

15. The order of the differential equation of all circles of given radius a is:

(A) 1

(B) 2

(C) 3

(D) 4

Ans: Correct option (B) because Let the equation of given family be $(x-h)^2 + (y-k)^2 = a^2$ It has two arbitrary constants h, k . Therefore, the order of the given differential equation will be two

16. The solution of the differential equation $2x \frac{dy}{dx} - y = 3$ represents a family of

(A) straight lines

(B) circles

(C) parabolas

(D) ellipses

Ans: Correct option (C) because Given equation can be written as

$$\frac{2 dy}{y+3} = \frac{dx}{x}$$

$$\Rightarrow 2 \log(y+3) = \log x + \log c$$

$$\Rightarrow (y+3)^2 = cx$$

which represents the family of parabolas

17. The integrating factor of the differential equation $\frac{dy}{dx}(x \log x) + y = 2 \log x$ is

(A) e^x

(B) $\log x$

(C) $\log(\log x)$

(D) x

Ans: Correct option (B) because given equation can be written as $\frac{dy}{dx} + \frac{y}{x \log x} = \frac{2}{x}$

Therefore, I.F. = $e^{\int \frac{1}{x \log x} dx} = e^{\log(\log x)} = \log x$

18. A solution of the differential equation $\left(\frac{dy}{dx}\right)^2 - x \frac{dy}{dx} + y = 0$ is

(A) $y = 2$

(B) $y = 2x$

(C) $y = 2x - 4$

(D) $y = 2x^2 - 4$

Ans: Correct option (C)

19. Which of the following is not a homogeneous function of x, y

(A) $x^2 + 2xy$

(B) $2x - y$

(C) $\cos^2\left(\frac{y}{x}\right) + \frac{y}{x}$

(D) $\sin x - \cos y$

Ans: Correct option (D)

20. Solution of the differential equation $\frac{dx}{x} + \frac{dy}{y} = 0$ is

(A) $\frac{1}{x} + \frac{1}{y} = c$

(B) $\log x \times \log y = c$

(C) $xy = c$

(D) $x + y = c$

Ans: Correct option (C) because from the given equation, we get $\log x + \log y = \log c$ giving $xy = c$

21. The solution of the differential equation $x \frac{dy}{dx} + 2y = x^2$ is

(A) $y = \frac{x^2 + c}{4x^2}$

(B) $y = \frac{x^2}{4} + c$

(C) $y = \frac{x^4 + c}{x^2}$

(D) $y = \frac{x^4 + c}{4x^2}$

Ans: Correct option (D) because I.F. $= e^{\int \frac{2}{x} dx} = e^{2 \log x} = e^{\log x^2} = x^2$

So, $y \times x^2 = \int x^2 x dx = \frac{x^4}{4} + k$

i.e. $y = \frac{x^4 + c}{4x^2}$

22. Fill in the blanks of the following:

(i) Order of the differential equation representing the family of parabolas $y^2 = 4ax$ is

Ans: One because a is the only arbitrary constant.

(ii) The degree of the differential equation $\left(\frac{dy}{dx}\right)^3 + \left(\frac{d^2y}{dx^2}\right)^2 = 0$ is

Ans: Two because degree of the highest order derivative is two.

(iii) The number of arbitrary constants in a particular solution of the differential equation $\tan x dx + \tan y dy = 0$ is

Ans: Zero because any particular solution of a differential equation has no arbitrary constant.

(iv) $F(x, y) = \frac{\sqrt{x^2 + y^2} + y}{x}$ is a homogeneous function of degree

Ans: Zero

(v) An appropriate substitution to solve the differential equation $\frac{dx}{dy} = \frac{x^2 \log\left(\frac{x}{y}\right)}{xy \log\left(\frac{x}{y}\right)}$

is

Ans: $x = vy$

(vi) Integrating factor of the differential equation $x \frac{dy}{dx} - y = \sin x$ is

Ans: $\frac{1}{x}$; given differential equation can be written as $\frac{dy}{dx} - \frac{y}{x} = \frac{\sin x}{x}$ and therefore

$$\text{I.F.} = e^{\int -\frac{1}{x} = e^{-\log x}} = \frac{1}{x}$$

(vii) The general solution of the differential equation $\frac{dy}{dx} = e^{x-y}$ is

Ans: $e^y = e^x + c$ from given equation, we have $e^y dy = e^x dx$

(viii) The general solution of the differential equation $\frac{dy}{dx} + \frac{y}{x} = 1$ is

Ans: $xy = \frac{x^2}{2} + c$; I.F. $= e^{\int \frac{1}{x} = e^{\log x} = x$ and the solution is $yx = \int x \cdot 1 dx = \frac{x^2}{2} + C$

(ix) The differential equation representing the family of curves $y = a \sin x + b \cos x$

Ans: $\frac{d^2y}{dx^2} + y = 0$

Differentiate with respect to x we get

$$\frac{dy}{dx} = A \cos x - B \sin x$$

$$\Rightarrow \frac{d^2y}{dx^2} = -A \sin x - B \cos x$$

$$\Rightarrow \frac{d^2y}{dx^2} + y = 0$$

(x) $\left(\frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}} \right) \frac{dx}{dy} = 1$ (x¹ 0) when written in the form $\frac{dy}{dx} + Py = Q$, then $P =$

Ans: $\frac{1}{\sqrt{x}}$, given equation can be written as

$$\frac{dy}{dx} = \frac{e^{-2\sqrt{x}}}{\sqrt{x}} - \frac{y}{\sqrt{x}}$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{\sqrt{x}} = \frac{e^{-2\sqrt{x}}}{\sqrt{x}}$$

This is a differential equation of the type $\frac{dy}{dx} + Py = Q$

23. State whether the following statements are True or False.

(i) Order of the differential equation representing the family of ellipses having centre at origin and foci on x -axis is two.

Ans: True because equation represents the given family $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, which has two arbitrary constants.

(ii) Degree of the differential equation $\sqrt{1 + \frac{d^2y}{dx^2}} = x + \frac{dy}{dx}$ is not defined

Ans: True because it is not a polynomial equation in its derivatives.

(iii) $\frac{dy}{dx} + y = 5$ is a differential equation of the type $\frac{dy}{dx} + Py = Q$ but it can be solved using variable separable method also.

Ans: True

(iv) $F(x, y) = \frac{y \cos\left(\frac{y}{x}\right) + x}{x \cos\left(\frac{y}{x}\right)}$ is not a homogeneous function.

Ans: True because $f(\lambda x, \lambda y) = \lambda^0 f(x, y)$

(v) $F(x, y) = \frac{x^2 + y^2}{x - y}$ is a homogeneous function of degree one

Ans: True because $f(\lambda x, \lambda y) = \lambda^1 f(x, y)$

(vi) Integrating factor of the differential equation $\frac{dy}{dx} - y = \cos x$ is e^x

Ans: False because I.F. = $e^{\int -1 dx} = e^{-x}$

(vii) The general solution of the differential equation $x(1+y^2)dx + y(1+x^2)dy = 0$ is $(1+x^2)(1+y^2) = k$

Ans: True because given equation can be written as

$$\frac{2x}{1+x^2}dx = \frac{-2y}{1+y^2}dy$$

$$\Rightarrow \log(1+x^2) = -\log(1+y^2) + \log k$$

$$\Rightarrow (1+x^2)(1+y^2) = k$$

(viii) The general solution of the differential equation $\frac{dy}{dx} + y \sec x = \tan x$ is $y(\sec x - \tan x) = \sec x - \tan x + x + k$

Ans: False because I.F. $= e^{\int \sec x dx} = e^{\log(\sec x + \tan x)} = \sec x + \tan x$

$$y(\sec x + \tan x) = \int (\sec x + \tan x) \tan x dx = \int (\sec x \tan x + \sec^2 x - 1) dx = \sec x + \tan x - x + k$$

(ix) $x + y = \tan^{-1}y$ is a solution of the differential equation $y^2 \frac{dy}{dx} + y^2 + 1 = 0$

Ans: True because $x + y = \tan^{-1}y$

$$1 + \frac{dy}{dx} = \frac{1}{1+y^2} \frac{dy}{dx}$$

$$\Rightarrow \frac{dy}{dx} \left(\frac{1}{1+y^2} - 1 \right) = 1$$

$$\Rightarrow \frac{dy}{dx} = \frac{-(1+y^2)}{y^2}$$

(x) $y = x$ is a particular solution of the differential equation $\frac{d^2y}{dx^2} - x^2 \frac{dy}{dx} + xy = x$

Ans: False, $y = x$ because does not satisfy the given differential equation

Exercise

1. Find the solution of $\frac{dy}{dx} = 2^{y-x}$

Ans: We have $\frac{dy}{dx} = 2^{y-x}$

$$\frac{dy}{dx} = \frac{2^y}{2^x} \quad \left[\because a^{m-n} = \frac{a^m}{a^n} \right]$$

$$\Rightarrow \frac{dy}{2^y} = \frac{dx}{2^x}$$

Integrate both sides:

$$\int 2^{-y} dy = \int 2^{-x} dx$$

$$\Rightarrow \frac{-2^{-y}}{\log 2} = \frac{-2^{-x}}{\log 2} + c$$

$$\Rightarrow -2^{-y} + 2^{-x} = + C \log 2$$

$$\Rightarrow 2^{\frac{1}{2}-y} - 2^{-y} = -C \log 2$$

$$\Rightarrow 2^{-x} - 2^{-y} = k \quad [\text{where } k = + C \log 2]$$

2. Find the differential equation of all non-vertical lines in a plane.

Ans: Since, the family of all non-vertical line is $y = mx + c$ where $m \neq \tan \frac{\pi}{2}$

On differentiating with respect to x we get:

$$\frac{dy}{dx} = m$$

$$\Rightarrow \frac{d^2y}{dx^2} = 0$$

3. Given that $\frac{dy}{dx} = e^{-2y}$ and $y = 0$ when $x = 5$. Find the value of x when $y = 3$

Ans: We have $\frac{dy}{dx} = e^{-2y}$

$$\frac{dy}{e^{-2y}} = dx$$

$$\Rightarrow \int e^{2y} dy = \int dx$$

$$\Rightarrow \frac{e^{2y}}{2} = x + C$$

When $x = 5, y = 0$ then substitute we get

$$\frac{e^0}{2} = 5 + C$$

$$\Rightarrow \frac{1}{2} = 5 + C$$

$$\Rightarrow C = \frac{1}{2} - 5 = -\frac{9}{2}$$

which implies $e^{2y} = 2x - 9$

when $y = 3$ then

$$e^6 = 2x - 9$$

$$\Rightarrow 2x = e^6 + 9$$

$$\Rightarrow x = \frac{(e^6 + 9)}{2}$$

4. Solve the differential equation $(x^2 - 1)\frac{dy}{dx} + 2xy = \frac{1}{x^2 - 1}$

Ans: We have $(x^2 - 1)\frac{dy}{dx} + 2xy = \frac{1}{x^2 - 1}$

Divide by $(x^2 - 1)$ on both sides

$$\frac{dy}{dx} + \left(\frac{2x}{x^2 - 1}\right)y = \frac{1}{(x^2 - 1)^2} \text{ which is a differential equation}$$

On comparing with $\frac{dy}{dx} + Py = Q$ we have

$$P = \frac{2x}{x^2-1}, Q = \frac{1}{(x^2-1)^2}$$

$$IF = e^{\int \frac{2x}{x^2-1} dx} = e^{\int \frac{2x}{x^2-1} dx}$$

$$\text{Put } x^2-1 = t$$

$$2x dx = dt$$

$$IF = e^{\int \frac{1}{t} dt} = e^{\log t} = t = (x^2-1)$$

Complete solution is

$$y \times IF = \int Q IF dx + k$$

$$\Rightarrow y \times (x^2-1) = \int \frac{1}{(x^2-1)^2} \times (x^2-1) dx + k$$

$$\Rightarrow y \times (x^2-1) = \int \frac{dx}{(x^2-1)}$$

$$\Rightarrow y \times (x^2-1) = \frac{1}{2} \log \left(\frac{x-1}{x+1} \right) + k$$

5. Solve the differential equation $\frac{dy}{dx} + 2xy = y$

Ans: we have $\frac{dy}{dx} + 2xy = y$

$$\frac{dy}{dx} + 2xy - y = 0$$

$$\Rightarrow \frac{dy}{dx} + (2x-1)y = 0$$

Now, compare with $\frac{dy}{dx} + Py = Q$ we get:

$$P = (2x-1), Q = 0$$

$$IF = e^{\int p dx} = e^{\int (2x-1) dx} = e^{\left(\frac{2x^2}{2} - x\right)} = e^{x^2 - x}$$

Complete solution is

$$\Rightarrow y \times e^{x^2 - x} = \int Q \times e^{x^2 - x} = 0 + C$$

$$\Rightarrow y \times e^{x^2 - x} = 0 + C$$

$$\Rightarrow y = Ce^{x - x^2}$$

6. Find the general solution of $\frac{dy}{dx} + ay = e^{xz}$

Ans: We have $\frac{dy}{dx} + ay = e^{xz}$

Now, compare with $\frac{dy}{dx} + Py = Q$ we get:

$$P = a, Q = e^{xz}$$

$$IF = e^{\int P dx} = e^{\int a dx} = e^{ax}$$

The general solution is

$$y \times e^{ax} = \int e^{mx} \times e^{ax} dx + C$$

$$\Rightarrow y \times e^{ax} = \frac{e^{(m+a)x}}{(m+a)} + C$$

$$\Rightarrow (m+a)y = \frac{e^{(m+a)x}}{e^{ax}} + \frac{(m+a)C}{e^{ax}}$$

$$\Rightarrow (m+a)y = \frac{e^{(m+ax)}}{e^{ax}} + \frac{(m+a)C}{e^{ax}}$$

$$\Rightarrow (m+a)y = e^{ax} + Ke^{-ax} \quad [\text{as } k = (m+a)C]$$

7. Solve the differential equation $\frac{dy}{dx} + 1 = e^{x+y}$

Ans: We have $\frac{dy}{dx} + 1 = e^{x+y}$ (i)

Substitute $x+y = t$ we get

$$1 + \frac{dy}{dx} = \frac{dt}{dx}$$

From (i), we have $\frac{dt}{dx} = e^t$

$$e^{-1} dt = dx$$

$$\Rightarrow -e^{-t} = x + C$$

$$\Rightarrow \frac{-1}{e^{x+y}} = x + C$$

$$\Rightarrow -1 = (x + C)e^{x+y}$$

$$\Rightarrow (x + C)e^{x+y} + 1 = 0$$

8. Solve $y dx - x dy = x^2 y dx$

Ans: We have $y dx - x dy = x^2 y dx$

$$\frac{1}{x^2} - \frac{1}{xy} \times \frac{dy}{dx} = 1$$

$$\Rightarrow -\frac{1}{xy} \times \frac{dy}{dx} + \frac{1}{x^2} - 1 = 0$$

$$\Rightarrow \frac{dy}{dx} - \frac{xy}{x^2} + xy = 0$$

$$\Rightarrow \frac{dy}{dx} - \frac{y}{x} + xy = 0$$

$$\Rightarrow \frac{dy}{dx} + \left(x - \frac{1}{x}\right)y = 0$$

Now, compare with $\frac{dy}{dx} + Py = Q$ we get:

$$P = \left(x - \frac{1}{x} \right), Q = 0$$

$$IF = e^{\int P dx} = e^{\int \left(x - \frac{1}{x} \right) dx} = e^{\frac{x^2}{2} - \log x} = e^{\frac{x^2}{2}} e^{-\log x} = e^{\frac{x^2}{2}} \times e^{-\log x} = \frac{1}{x} e^{\frac{x^2}{2}}$$

General solution is

$$y \times \frac{1}{x} e^{\frac{x^2}{2}} = \int 0 \times \frac{1}{x} e^{\frac{x^2}{2}} dx + C$$

$$\Rightarrow y \times \frac{1}{x} e^{\frac{x^2}{2}} = C$$

$$\Rightarrow y = C x e^{\frac{x^2}{2}}$$

9. Solve the differential equation $\frac{dy}{dx} = 1 + x + y^2 + xy^2$ when $y = 0, x = 0$

Ans: Given $\frac{dy}{dx} = 1 + x + y^2 + xy^2$

$$\frac{dy}{dx} = (1 + x) + y^2(1 + x)$$

$$\Rightarrow \frac{dy}{dx} = (1 + y^2)(1 + x)$$

$$\Rightarrow \frac{dy}{1 + y^2} = (1 + x) dx$$

Now, integrate both sides

$$\Rightarrow \tan^{-1} y = x + \frac{x^2}{2} + K$$

When $y = 0, x = 0$ then substitute in above equation, we get

$$\Rightarrow \tan^{-1}(0) = 0 + 0 + K$$

$$\Rightarrow K = 0$$

$$\Rightarrow \tan^{-1}y = x + \frac{x^2}{2}$$

$$\Rightarrow y = \tan\left(x + \frac{x^2}{2}\right)$$

10. Find the general solution of $(x + 2y^3)\frac{dy}{dx} = y$

Ans: We have $(x + 2y^3)\frac{dy}{dx} = y$

$$y \frac{dx}{dy} = x + 2y^3$$

$$\Rightarrow \frac{dx}{dy} = \frac{x}{y} + 2y^2$$

which is a differential equation

Now, compare with $\frac{dy}{dx} + Py = Q$ we get:

$$P = -\frac{1}{y}, Q = 2y^2$$

$$\text{IF} = e^{\int -\frac{1}{y} dy} = e^{-\int \frac{1}{y} dy} = e^{-\log y} = \frac{1}{y}$$

Complete solution is

$$x \times \text{I.F.} = \int Q \times \text{I.F.} dy + C$$

$$\text{General solution is } x \frac{1}{y} = \int 2y^2 \times \frac{1}{y} dy + C$$

$$\frac{x}{y} = \frac{2y^2}{2} + C$$

$$\Rightarrow \frac{x}{y} = y^2 + C$$

$$\Rightarrow x = y^3 + Cy$$

11. If $y(x)$ is a solution of $\left(\frac{2+\sin x}{1+y}\right)\frac{dy}{dx} = -\cos x$, $y(0) = 1$ then find value of $y\left(\frac{\pi}{2}\right)$

Ans: We have $\left(\frac{2+\sin x}{1+y}\right)\frac{dy}{dx} = -\cos x$

$$\frac{dy}{1+y} = -\frac{\cos x}{2+\sin x} dx$$

Integrate both sides:

$$\int \frac{1}{1+y} dy = -\int \frac{\cos x}{2+\sin x} dx - \int \frac{\cos x}{2+\sin x} dx$$

$$\Rightarrow \log(1+y) = -\log(2+\sin x) + \log C$$

$$\Rightarrow \log(1+y) + \log(2+\sin x) = \log C$$

$$\Rightarrow \log(1+y)(2+\sin x) = \log C$$

$$\Rightarrow (1+y)(2+\sin x) = C$$

$$\Rightarrow 1+y = \frac{C}{2+\sin x}$$

$$\Rightarrow y = \frac{C}{2+\sin x} - 1$$

When $x = 0$, $y = 1$ then $C = 4$

Put $C = 4$ in above equation, we have:

$$Y = \frac{4}{2+\sin x} - 1$$

$$\Rightarrow Y\left(\frac{\pi}{2}\right) = \frac{4}{2+\sin \frac{\pi}{2}} - 1 = \frac{4}{2+1} - 1$$

$$= \frac{4}{3} - 1 = \frac{1}{3}$$

12. If $y(t)$ is a solution of $(1+t)\frac{dy}{dt} - ty = 1$, $y(0) = -1$ then show that $y(1) = -\frac{1}{2}$

Ans: We have $(1+t)\frac{dy}{dt} - ty = 1$

Divide both sides by $(1+t)$

$\frac{dy}{dt} - \left(\frac{t}{1+t}\right)y = \frac{1}{1+t}$ which is a linear differential equation

Compare with $\frac{dy}{dt} + Py = Q$ we get

$$P = -\left(\frac{t}{1+t}\right), Q = \frac{1}{1+t}$$

$$IF = e^{-\int \frac{t}{1+t} dt} = e^{-\int \left(1 - \frac{1}{1+t}\right) dt} = e^{-t} \times e^{\log(1+t)} = e^{-t}(1+t)$$

Complete solution is $y \times I.F. = \int Q \times I.F. dx + C$

$$\text{General solution is } y(t) \times \frac{(1+t)}{e^t} = \int \frac{(1+t) \times e^{-1}}{(1+t)} dt + C y(t) \times \frac{(1+t)}{e^t} = \int \frac{(1+t) \times e^{-t}}{(1+t)} dt + C$$

$$y(t) = \frac{e^{-t}}{(-1)} \times \frac{e^t}{1+t} + C' \text{ where } C' = \frac{Ce^t}{1+t}$$

$$\Rightarrow y(t) = \frac{e^{-t}}{(-1)} \times \frac{e^t}{1+t} + C'$$

$$\Rightarrow y(t) = -\frac{1}{1+t} + C'$$

When $t = 0, y = -1$ then

$$-1 = -1 + C'$$

$$\Rightarrow C' = 0$$

$$\therefore y(t) = -\frac{1}{1+t}$$

$$\Rightarrow y(1) = -\frac{1}{2}$$

13. Form the differential equation having $y = (\sin^{-1}x)^2 + A\cos^{-1}x + B$ where A and B are arbitrary constants as its general solution

Ans: We have $y = (\sin^{-1}x)^2 + A\cos^{-1}x + B$

Now, differentiate we get:

$$\frac{dy}{dx} = \frac{2\sin^{-1}x}{\sqrt{1-x^2}} + \frac{(-A)}{\sqrt{1-x^2}}$$

$$\Rightarrow \sqrt{1-x^2} \frac{dy}{dx} = 2\sin^{-1}x - A$$

Again, differentiate we get:

$$\sqrt{1-x^2} \frac{d^2y}{dx^2} + \frac{dy}{dx} \times \frac{-2x}{2\sqrt{1-x^2}} = \frac{2}{\sqrt{1-x^2}}$$

$$\Rightarrow (1-x^2) \frac{d^2y}{dx^2} - \frac{x}{\sqrt{1-x^2}} \times \sqrt{1-x^2} \frac{dy}{dx} = 2$$

$$\Rightarrow (1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} = 2$$

$$\Rightarrow (1-x^2) \frac{d^2y}{dx^2} - x \frac{dy}{dx} - 2 = 0$$

14. Form the differential equation of all circles which pass through origin and whose centres lie on y - axis

Ans: We have given that circles pass through origin and their centre lie on y - axis

Let (0, k) be the centre of circle with radius k

So, equation of circle is

$$(x - 0)^2 + (y - k)^2 = k^2$$

$$\Rightarrow x^2 + (y - k)^2 = k^2$$

$$\Rightarrow x^2 + y^2 - 2ky = 0$$

$$\Rightarrow \frac{x^2 + y^2}{2y} = k$$

Now, differentiate we get:

$$\frac{2y \left(2x + 2y \frac{dy}{dx} \right) - (x^2 + y^2) \frac{2dy}{dx}}{4y^2} = 0$$

$$\Rightarrow 4y \left(x + y \frac{dy}{dx} \right) - 2(x^2 + y^2) \frac{dy}{dx} = 0$$

$$\Rightarrow 4xy + 4y^2 \frac{dy}{dx} - 2(x^2 + y^2) \frac{dy}{dx} = 0$$

$$\Rightarrow [4y^2 - 2(x^2 + y^2)] \frac{dy}{dx} + 4xy = 0$$

$$\Rightarrow (4y^2 - 2x^2 - 2y^2) \frac{dy}{dx} + 4xy = 0$$

$$\Rightarrow (2y^2 - 2x^2) \frac{dy}{dx} + 4xy = 0$$

$$\Rightarrow (y^2 - x^2) \frac{dy}{dx} + 2xy = 0$$

$$\Rightarrow (x^2 - y^2) \frac{dy}{dx} - 2xy = 0$$

15. Find the equation of a curve passing through origin and satisfying the differential equation $(1 + x^2) \frac{dy}{dx} + 2xy = 4x^2$

Ans: We have $(1 + x^2) \frac{dy}{dx} + 2xy = 4x^2$

Divide both sides by $(1 + x^2)$

$$\frac{dy}{dx} + \frac{2x}{1+x^2} \times y = \frac{4x^2}{1+x^2} \text{ which is a differential equation}$$

Now, compare with $\frac{dy}{dx} + Py = Q$ we get:

$$P = \frac{2x}{1+x^2}, Q = \frac{4x^2}{1+x^2}$$

$$IF = e^{\int P dx} = e^{\int \frac{2x}{1+x^2} dx}$$

$$\text{Put } 1+x^2 = t \Rightarrow 2x dx = dt$$

$$\text{We have } IF = 1+x^2 = e^{\int \frac{dt}{t}} = e^{\log t} = e^{\log(1+x^2)}$$

The general solution is

$$y \times (1+x^2) = \int \frac{4x^2}{1+x^2} (1+x^2) dx + C$$

$$\Rightarrow y \times (1+x^2) = \int 4x^2 dx + C$$

$$\Rightarrow y \times (1+x^2) = 4 \frac{x^3}{3} + C$$

As curve passes through origin then substitute $x = 0, y = 0 \Rightarrow C = 0$

So, equation of curve is

$$y(1+x^2) = \frac{4x^3}{3}$$

$$\Rightarrow y = \frac{4x^3}{3(1+x^2)}$$

16. Solve $x^2 \frac{dy}{dx} = x^2 + xy + y^2$

Ans: We have $x^2 \frac{dy}{dx} = x^2 + xy + y^2$

Divide both side by x^2 we get $\frac{dy}{dx} = 1 + \frac{y}{x} + \frac{y^2}{x^2}$

Let $f(x,y) = 1 + \frac{y}{x} + \frac{y^2}{x^2}$

$$\Rightarrow f(\lambda x, \lambda y) = 1 + \frac{\lambda y}{\lambda x} + \frac{\lambda^2 y^2}{\lambda^2 x^2}$$

$$f(\lambda x, \lambda y) = \lambda^0 \left(1 + \frac{y}{x} + \frac{y^2}{x^2} \right) = \lambda^0 f(x,y) \text{ is a homogeneous equation of degree zero}$$

Put $y = vx \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$

Now, substitute we get

$$\left(v + x \frac{dv}{dx} \right) = 1 + v + v^2$$

$$\Rightarrow x \frac{dv}{dx} = 1 + v + v^2 - v$$

$$\Rightarrow x \frac{dv}{dx} = 1 + v^2$$

$$\Rightarrow \frac{dv}{1+v^2} = \frac{dx}{x}$$

Integrate both sides, we get

$$\int \frac{dv}{1+v^2} = \int \frac{dx}{x}$$

$$\Rightarrow \tan^{-1} v = \log|x| + C$$

$$\Rightarrow \tan^{-1} \left(\frac{y}{x} \right) = \log|x| + C$$

17. Find the general solution of the differential equation $(1+y^2) + (x - e^{\tan^{-1}y}) \frac{dy}{dx} = 0$

Ans: We have $(1+y^2) + (x - e^{\tan^{-1}y}) \frac{dy}{dx} = 0$

$$\Rightarrow (1+y^2) = -\left(x - e^{\tan^{-1}y}\right) \frac{dy}{dx}$$

$$\Rightarrow (1+y^2) \frac{dx}{dy} = -x + e^{\tan^{-1}y}$$

$$\Rightarrow (1+y^2) \frac{dx}{dy} + x = e^{\tan^{-1}y}$$

$$\Rightarrow \frac{dx}{dy} + \frac{x}{1+y^2} = \frac{e^{\tan^{-1}y}}{1+y^2}$$

Now, compare with $\frac{dx}{dy} + Px = Q$ we get

$$P = \frac{1}{1+y^2}, Q = \frac{e^{\tan^{-1}y}}{1+y^2}$$

$$IF = e^{\int \frac{1}{1+y^2} dy} = e^{\tan^{-1}y}$$

General solution is

$$xe^{\tan^{-1}y} = \int \frac{e^{\tan^{-1}y}}{1+y^2} e^{\tan^{-1}y} dy + C$$

$$\Rightarrow xe^{\tan^{-1}y} = \int \frac{(e^{\tan^{-1}y})^2}{1+y^2} dy + C$$

Put $\tan^{-1}y = t \Rightarrow \frac{1}{1+y^2} dy = dt$ we have

$$x \times e^{\tan^{-1}y} = \int e^{2t} dt + C$$

$$\Rightarrow x \times e^{\tan^{-1}y} = \frac{1}{2} e^{2\tan^{-1}y} + C$$

$$\Rightarrow 2xe^{\tan^{-1}y} = e^{2\tan^{-1}y} + 2C$$

$$\Rightarrow 2xe^{\tan^{-1}y} = e^{2\tan^{-1}y} + K$$

18. Find the general solution of $y^2 dx + (x^2 - xy + y^2) dy = 0$

Ans: We have $y^2 dx + (x^2 - xy + y^2) dy = 0$

$$y^2 dx = -(x^2 - xy + y^2) dy$$

$$\Rightarrow y^2 \frac{dx}{dy} = -(x^2 - xy + y^2)$$

Now, divide by y^2 on both sides, we get:

$$\frac{dx}{dy} = -\left(\frac{x^2}{y^2} - \frac{x}{y} + 1\right) \text{ is homogeneous equation}$$

Put $\frac{x}{y} = v$ or $x = vy$ we have

$$\frac{dx}{dy} = v + y \frac{dv}{dy}$$

Now, substitute we get:

$$v + y \frac{dv}{dy} = -[v^2 - v + 1]$$

$$\Rightarrow y \frac{dv}{dy} = -v^2 + v - 1 - v$$

$$\Rightarrow y \frac{dv}{dy} = -v^2 - 1$$

$$\Rightarrow \frac{dv}{v^2 + 1} = -\frac{dy}{y}$$

Now, integrate

$$\int \frac{dv}{v^2 + 1} = \int -\frac{dy}{y}$$

$$\Rightarrow \tan^{-1}(v) = -\log y + C$$

$$\Rightarrow \tan^{-1}\left(\frac{x}{y}\right) + \log y = C$$

19. Solve $(x + y)(dx - dy) = dx + dy$ [Hint: substitute $x + y = z$ after separating dx, dy]

Ans: We have $(x+y)(dx-dy) = dx+dy$

$$(x+y)\left(1-\frac{dy}{dx}\right) = 1+\frac{dy}{dx}$$

Put $x+y = z$ we have $1+\frac{dy}{dx} = \frac{dz}{dx}$

Substitute these values we get:

$$z\left(1-\frac{dz}{dx}+1\right) = \frac{dz}{dx}$$

$$\Rightarrow z\left(2-\frac{dz}{dx}\right) = \frac{dz}{dx}$$

$$\Rightarrow 2z-z\frac{dz}{dx}-\frac{dz}{dx} = 0$$

$$\Rightarrow 2z-(z+1)\frac{dz}{dx} = 0$$

$$\Rightarrow \frac{dz}{dx} = \frac{2z}{z+1}$$

$$\Rightarrow \left(\frac{z+1}{z}\right)dz = 2 dx$$

Integrate both sides, we get:

$$\int \left(1+\frac{1}{z}\right)dz = 2 \int dx$$

$$\Rightarrow z+\log z = 2x-\log C$$

$$\Rightarrow (x+y)+\log(x+y) = 2x-\log C$$

$$\Rightarrow 2x-x-y = \log C+\log(x+y)$$

$$\Rightarrow x-y = \log|C(x+y)|$$

$$\Rightarrow e^{x-y} = C(x+y)$$

$$\Rightarrow (x+y) = \frac{1}{C}e^{x-y}$$

$$\Rightarrow x+y = Ke^{x-y}$$

20. Solve $2(y+3) - xy \frac{dy}{dx} = 0$ given that $y(1) = -2$

Ans: We have $2(y+3) - xy \frac{dy}{dx} = 0$

$$2(y+3) = xy \frac{dy}{dx}$$

$$\Rightarrow 2 \frac{dx}{x} = \left(\frac{y}{y+3} \right) dy$$

$$\Rightarrow 2 \times \frac{dx}{x} = \left(\frac{y+3-3}{y+3} \right) dy$$

$$\Rightarrow 2 \times \frac{dx}{x} = \left(1 - \frac{3}{y+3} \right) dy$$

Now, integrate both sides:

$$\int 2 \frac{dx}{x} = \int \left(1 - \frac{3}{y+3} \right) dy$$

$$\Rightarrow 2 \log x = y - 3 \log(y+3) + C$$

When $x = 1$, $y = -2$ then

$$2 \log 1 = -2 - 3 \log(-2+3) + C$$

$$\Rightarrow 2(0) = -2 - 3(0) + C$$

$$\Rightarrow C = 2$$

Now, substitute we get

$$2 \log x = y - 3 \log(y+3) + 2$$

$$\Rightarrow 2 \log x + 3 \log(y+3) = y + 2$$

$$\Rightarrow \log x^2 + \log(y+3)^3 = (y+2)$$

$$\Rightarrow \log x^2 (y+3)^3 = y+2$$

$$\Rightarrow x^2 (y+3)^3 = e^{y+2}$$

21. Solve the differential equation $dy = \cos x(2 - y \operatorname{cosec} x)dx$ given that $x = \frac{\pi}{2}$

Ans: We have $dy = \cos x(2 - y \operatorname{cosec} x)dx$

$$\frac{dy}{dx} = \cos x (2 - y \operatorname{cosec} x)$$

$$\Rightarrow \frac{dy}{dx} = 2 \cos x - y \operatorname{cosec} x \cos x$$

$$\Rightarrow \frac{dy}{dx} = 2 \cos x - y \cot x$$

$$\Rightarrow \frac{dy}{dx} + y \cot x = 2 \cos x$$

which is a differential equation

Now, compare with $\frac{dy}{dx} + Py = Q$ we get

$$P = \cot x, Q = 2 \cos x$$

$$IF = e^{\int P dx} = e^{\int \cot x dx} = e^{\log \sin x} = \sin x$$

General solution

$$y \times \sin x = \int 2 \cos x \sin x dx + C$$

$$\Rightarrow y \times \sin x = \int \sin 2x dx + C \quad [\sin 2x = 2 \sin x \cos x]$$

$$\Rightarrow y \times \sin x = -\frac{\cos 2x}{2} + C$$

When $x = \frac{\pi}{2}$, $y = 2$ then

$$2 \times \sin \frac{\pi}{2} = -\frac{\cos \left(2 \times \frac{\pi}{2} \right)}{2} + C$$

$$\Rightarrow 2 \times 1 = +\frac{1}{2} + C$$

$$\Rightarrow 2 - \frac{1}{2} = C$$

$$\Rightarrow \frac{4-1}{2} = C$$

$$\Rightarrow C = \frac{3}{2}$$

Now, substitute values we get:

$$y \sin x = -\frac{1}{2} \cos 2x + \frac{3}{2}$$

22. Form the differential equation by eliminating A and B in $Ax^2 + By^2 = 1$

Ans: We have $Ax^2 + By^2 = 1$

Differentiate with respect to x we get:

$$2Ax + 2By \frac{dy}{dx} = 0$$

$$\Rightarrow 2By \frac{dy}{dx} = -2Ax$$

$$By \frac{dy}{dx} = -Ax$$

$$\Rightarrow \frac{y}{x} \frac{dy}{dx} = -\frac{A}{B}$$

Now, again differentiate with respect to x we get:

$$\frac{y}{x} \times \frac{d^2y}{dx^2} + \frac{dy}{dx} \times \left(\frac{x \frac{dy}{dx} - y}{x^2} \right) = 0$$

$$\Rightarrow \frac{y}{x} \times \frac{d^2y}{dx^2} + \frac{x \left(\frac{dy}{dx} \right)^2 - y \left(\frac{dy}{dx} \right)}{x^2} = 0$$

$$\Rightarrow xy \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 - y \left(\frac{dy}{dx} \right) = 0$$

$$\Rightarrow xyy'' + x(y')^2 - yy' = 0$$

23. Solve the differential equation $(1+y^2) \tan^{-1}x dx + 2y(1+x^2) dy = 0$

Ans: We have $(1+y^2) \tan^{-1}x dx + 2y(1+x^2) dy = 0$

$$(1+y^2)\tan^{-1}x dx = -2y(1+x^2)dy$$

$$\Rightarrow \frac{\tan^{-1}x dx}{1+x^2} = -\frac{2y}{1+y^2} dy$$

Integrate on both sides:

$$\Rightarrow \int \frac{\tan^{-1}x}{1+x^2} dx = -\int \frac{2y}{1+y^2} dy$$

Put $\tan^{-1}x = t$ in LHS we get $\frac{1}{1+x^2} dx = dt$

Put $1+y^2 = u$ in RHS we get:

$$2y dy = du$$

$$\Rightarrow \int t dt = -\int \frac{1}{u} du$$

$$\Rightarrow \frac{t^2}{2} = -\log u + C$$

$$\Rightarrow \frac{1}{2}(\tan^{-1}x)^2 = -\log(1+y^2) + C$$

$$\Rightarrow \frac{1}{2}(\tan^{-1}x)^2 + \log(1+y^2) = C$$

24. Find the differential equation of system of concentric circles with centre (1,2)

Ans: The family of concentric circles with centre (1,2) and radius a is given by

$$(x-1)^2 + (y-2)^2 = a^2$$

$$\Rightarrow x^2 + 1 - 2x + y^2 + 4 - 4y = a^2$$

$$\Rightarrow x^2 + y^2 - 2x - 4y + 5 = a^2$$

Now differentiate with respect to x we get:

$$2x+2y \frac{dy}{dx} - 2 - 4 \frac{dy}{dx} = 0$$

$$\Rightarrow (2x-4) \frac{dy}{dx} + 2x-2 = 0$$

$$\Rightarrow (y-2) \frac{dy}{dx} + (x-1) = 0$$

Long Answer Questions

25. Solve $y + \frac{d}{dx}(xy) = x(\sin x + \log x)$

Ans: We have $y + \frac{d}{dx}(xy) = x(\sin x + \log x)$

$$y + x \frac{d}{dx} y + y = x(\sin x + \log x) y + x \frac{dy}{dx} + y = x(\sin x + \log x)$$

$$\Rightarrow x \frac{dy}{dx} + 2y = x(\sin x + \log x)$$

Divide by x on both sides we get:

$$\frac{dy}{dx} + \frac{2}{x} y = \sin x + \log x \text{ is a linear differential equation}$$

Now, compare with $\frac{dy}{dx} + Px = Q$ we get

$$P = \frac{2}{x}, Q = \sin x + \log x$$

$$IF = e^{\int \frac{2}{x} dx} = e^{2 \log x} = x^2$$

Complete solution is

$$y \times IF = \int Q \times IF dx + C$$

General solution is

$$yx^2 = \int (\sin x + \log x)x^2 dx + C$$

$$\Rightarrow yx^2 = \int (x^2 \sin x + x^2 \log x) dx + C$$

$$\Rightarrow yx^2 = \int x^2 \sin x dx + \int x^2 \log x dx + C$$

$$\Rightarrow yx^2 = I_1 + I_2 + C$$

Let $I_1 = \int x^2 \sin x dx$ we get

$$= x^2(-\cos x) + \int 2x \cos x dx$$

$$= -x^2 \cos x + \left[2x(\sin x) - \int 2 \sin x dx \right]$$

$$I_1 = -x^2 \cos x + 2x \sin x + 2 \cos x$$

Let $I_2 = \int x^2 \log x dx$ we get

$$= \log x \frac{x^3}{3} - \int \frac{1}{x} \frac{x^3}{3} dx$$

$$= \log x \frac{x^3}{3} - \frac{1}{3} \int x^2 dx$$

$$= \log x \frac{x^3}{3} - \frac{1}{3} \frac{x^3}{3}$$

Now, substitute value of I_1, I_2 we get

$$yx^2 = -x^2 \cos x + 2x \sin x + 2 \cos x + \frac{x^3}{3} \log x - \frac{1}{9} x^3 + C$$

$$\Rightarrow y = -\cos x + \frac{2 \sin x}{x} + \frac{2 \cos x}{x^2} + \frac{x}{3} \log x - \frac{x}{9} + Cx^{-2}$$

26. Find the general solution of $(1 + \tan y)(dx - dy) + 2x dy = 0$

Ans: We have $(1 + \tan y)(dx - dy) + 2x dy = 0$

Divide by dy we get:

$$(1+\tan y)\left(\frac{dx}{dy}-1\right)+2x=0$$

$$\Rightarrow (1+\tan y)\frac{dx}{dy}-(1+\tan y)+2x=0$$

$$\Rightarrow (1+\tan y)\frac{dx}{dy}+2x=(1+\tan y)$$

Divide by $(1+\tan y)$ on both sides we get:

$$\frac{dx}{dy}+\frac{2x}{1+\tan y}=1 \text{ is a linear differential equation}$$

Compare with $\frac{dx}{dy}+Px=Q$ we get

$$P=\frac{2}{1+\tan y}, Q=1$$

$$IF=e^{\int \frac{2}{1+\tan y} dy}=e^{\int \frac{2\cos y}{\cos y+\sin y} dy}$$

$$=e^{\int \frac{\cos y+\sin y+\cos y-\sin y}{\cos y+\sin y} dy}$$

$$=e^{\int (1+\frac{\cos y-\sin y}{\cos y+\sin y}) dy}$$

$$=e^{y+\log x(\cos y+\sin y)}$$

$$=e^y \times (\cos y+\sin y) \quad [e^{\log x}=x]$$

General solution is

$$x \times e^y (\cos y+\sin y)=\int 1 \times e^y (\cos y+\sin y) dy+C$$

$$\Rightarrow x \times e^y (\cos y+\sin y)=\int e^y (\sin y+\cos y) dy+C$$

$$\Rightarrow x \times e^y (\cos y+\sin y)=e^y \sin y+C \quad \left[\int e^x \{f(x)+f'(x)\} dx = e^x f(x) \right]$$

$$\Rightarrow x(\sin y+\cos y)=\sin y+Ce^{-y}$$

27. Solve $\frac{dy}{dx}=\cos(x+y)+\sin(x+y)$

Ans: We have $\frac{dy}{dx} = \cos(x+y) + \sin(x+y)$

Put $x+y = z$ we get $1 + \frac{dy}{dx} = \frac{dz}{dx}$

Substitute values we get

$$\begin{aligned} \left(\frac{dz}{dx} - 1 \right) &= \cos z + \sin z \\ \Rightarrow \frac{dz}{dx} &= (\cos z + \sin z + 1) \\ \Rightarrow \frac{dz}{\cos z + \sin z + 1} &= dx \end{aligned}$$

Integrate both sides we get:

$$\begin{aligned} \int \frac{dx}{\cos z + \sin z + 1} &= \int 1 dx \\ \Rightarrow \int \frac{dz}{\frac{1 - \tan^2 \frac{z}{2}}{1 + \tan^2 \frac{z}{2}} + \frac{2 \tan \frac{z}{2}}{1 + \tan^2 \frac{z}{2}} + 1} &= \int dx \\ \Rightarrow \int \frac{dz}{1 - \tan^2 \frac{z}{2} + 2 \tan \frac{z}{2} + 1 + \tan^2 \frac{z}{2}} &= \int dx \\ \Rightarrow \int \frac{\left(1 + \tan^2 \frac{z}{2} \right)}{2 + 2 \tan^2 \frac{z}{2}} &= \int dx \\ \Rightarrow \int \frac{\sec^2 \frac{z}{2} dz}{2(1 + \tan^2 \frac{z}{2})} &= \int dx \end{aligned}$$

Put $1 + \tan^2 \frac{z}{2} = t$ we get $\left(\frac{1}{2} \sec^2 \frac{z}{2} \right) dz = dt$

$$\int \frac{dt}{t} = \int dx$$

$$\Rightarrow \log |t| = x+C$$

$$\Rightarrow e^{i\theta} \log \left| 1 + \tan \frac{z}{2} \right| = x+C$$

$$\Rightarrow \log \left| 1 + \tan \frac{(x+y)}{2} \right| = x+C$$

28. Find the general solution of $\frac{dy}{dx} - 3y = \sin 2x$

Ans: We have $\frac{dy}{dx} - 3y = \sin 2x$ is a linear differential equation

Compare with $\frac{dy}{dx} + Py = Q$ we get

$$P = -3, Q = \sin 2x$$

$$IF = e^{-3 \int dx} = e^{-3x}$$

General solution is

$$ye^{-3x} = \int \sin 2x e^{-3x} dx$$

Let $ye^{-3x} = I$

$$I = \int e^{-3x} \sin 2x$$

$$\Rightarrow I = \sin 2x \left(\frac{e^{-3x}}{-3} \right) - \int 2 \cos 2x \left(\frac{e^{-3x}}{-3} \right) dx + C_1$$

$$\Rightarrow I = -\frac{1}{3} e^{-3x} \sin 2x + \frac{2}{3} \int e^{-3x} \cos 2x dx + C_1$$

$$\Rightarrow I = -\frac{1}{3} e^{-3x} \sin 2x + \frac{2}{3} \left(\cos 2x \frac{e^{-3x}}{-3} - \int (-2 \sin 2x) \frac{e^{-3x}}{-3} dx \right) + C_1 + C_2$$

$$\Rightarrow I = -\frac{1}{3} e^{-3x} \sin 2x - \frac{2}{9} \cos 2x e^{-3x} - \frac{4}{9} I + C'$$

$$\Rightarrow I + \frac{4I}{9} = +e^{-3x} \left(-\frac{1}{3} \sin 2x - \frac{2}{9} \cos 2x \right) + C'$$

$$\Rightarrow \frac{13}{9} I = e^{-3x} \left(-\frac{1}{3} \sin 2x - \frac{2}{9} \cos 2x \right) + C'$$

$$\Rightarrow I + \frac{4I}{9} = e^{-3x} \left(-\frac{1}{3} \sin 2x - \frac{2}{9} \cos 2x \right) + C'$$

$$\Rightarrow I = \frac{9}{13} e^{-3x} \left(-\frac{1}{3} \sin 2x - \frac{2}{9} \cos 2x \right) + C \quad \left[\text{where } C = \frac{9C'}{13} \right]$$

$$\Rightarrow I = \frac{3}{13} e^{-3x} \left(-\sin 2x - \frac{2}{3} \cos 2x \right) + C$$

$$\Rightarrow I = \frac{3}{13} e^{-3x} \frac{(-3 \sin 2x - 2 \cos 2x)}{3} + C$$

$$\Rightarrow \frac{e^{-3x}}{13} (-3 \sin 2x - 2 \cos 2x) + C$$

$$\Rightarrow I = \frac{e^{-3x}}{13} (2 \cos 2x + 3 \sin 2x) + C$$

Substitute values we get

$$y e^{-3x} = -\frac{e^{-3x}}{13} (2 \cos 2x + 3 \sin 2x) + C$$

$$\Rightarrow y = -\frac{1}{13} (2 \cos 2x + 3 \sin 2x) + C e^{3x}$$

29. Find the equation of a curve passing through (2,1) . if the slope of the tangent to the curve at any point (x,y) is $\frac{x^2 + y^2}{2xy}$

Ans: The slope of tangent to curve at point (x,y) is $\frac{x^2 + y^2}{2xy}$

$$\left(\frac{dy}{dx}\right)_{(x,y)} = \frac{x^2 + y^2}{2xy}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{2} \left(\frac{x}{y} + \frac{y}{x} \right)$$

Put $y = vx$ we get $\frac{dy}{dx} = v + x \frac{dv}{dx}$

Substitute values we get:

$$v + x \frac{dv}{dx} = \frac{1}{2} \left(\frac{1}{v} + v \right)$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{1}{2} \left(\frac{1+v^2}{v} \right)$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1+v^2}{2v} - v$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1+v^2-2v^2}{2v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{1-v^2}{2v}$$

$$\Rightarrow \frac{2v}{1-v^2} dv = \frac{dx}{x}$$

Integrate both sides we get:

$$\int \frac{2v}{1-v^2} dv = \int \frac{dx}{x}$$

Put $1-v^2 = t$ in LHS we get:

$$-2v dv = dt$$

$$\Rightarrow -\int \frac{dt}{t} = \int \frac{dx}{x}$$

$$\Rightarrow -\log t = \log x + \log C$$

$$\Rightarrow -\log(1-v^2) = \log x + \log C$$

$$\Rightarrow -\log\left(1 - \frac{y^2}{x^2}\right) = \log x + \log C$$

$$\Rightarrow -\log\left(\frac{x^2 - y^2}{x^2}\right) = \log x + \log C$$

$$\Rightarrow \log\left(\frac{x^2}{x^2 - y^2}\right) = \log x + \log C$$

$$\Rightarrow \frac{x^2}{x^2 - y^2} = Cx$$

As curve passes through point (2,1)

$$\frac{(2)^2}{(2)^2 - (1)^2} = C(2)$$

$$\Rightarrow C = \frac{2}{3}$$

Required solution is $2(x^2 - y^2) = 3x$

30. Find the equation of the curve through the point (1,0) if the slope of the tangent to the curve at any point (x,y) is $\frac{y-1}{x^2+x}$

Ans: It is given that, slope of tangent to the curve at any point (x,y) is $\frac{y-1}{x^2+x}$

$$\left(\frac{dy}{dx}\right)_{(x,y)} = \frac{y-1}{x^2+x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{y-1}{x^2+x}$$

$$\Rightarrow \frac{dy}{y-1} = \frac{dx}{x^2+x}$$

$$\Rightarrow \int \frac{dy}{y-1} = \int \frac{dx}{x^2+x}$$

$$\Rightarrow \int \frac{dy}{y-1} = \int \frac{dx}{x(x+1)}$$

$$\Rightarrow \int \frac{dy}{y-1} = \int \left(\frac{1}{x} - \frac{1}{x+1} \right) dx$$

$$\Rightarrow \log(y-1) = \log x - \log(x+1) + \log C$$

$$\Rightarrow \log(y-1) = \log \left(\frac{xC}{x+1} \right)$$

As given curve passes through point (1,0)

$$0-1 = \frac{1.C}{1+1}$$

$$\Rightarrow C = -2$$

Particular solution is

$$y-1 = \frac{-2x}{x+1}$$

$$\Rightarrow (y-1)(x+1) = -2x$$

$$\Rightarrow (y-1)(x+1) + 2x = 0$$

31. Find the equation of a curve passing through origin if the slope of the tangent to the curve at any point (x,y) is equal to the square of the difference of the abscissa and ordinate of the point.

Ans: Slope of tangent to the curve = $\frac{dy}{dx}$

and difference of abscissa and ordinate = $x-y$

According to the question $\frac{dy}{dx} = (x-y)^2$

Put $x-y = z$

$$1 - \frac{dy}{dx} = \frac{dz}{dx}$$

$$\Rightarrow \frac{dy}{dx} = 1 - \frac{dz}{dx}$$

Substitute values we get

$$1 - \frac{dz}{dx} = z^2$$

$$\Rightarrow 1 - z^2 = \frac{dz}{dx}$$

$$\Rightarrow dx = \frac{dz}{1 - z^2}$$

Integrate both sides, we get:

$$\int dx = \int \frac{dz}{1 - z^2}$$

$$\Rightarrow x = \frac{1}{2} \log \left| \frac{1+z}{1-z} \right| + C$$

$$\Rightarrow tx = \frac{1}{2} \log \left| \frac{1+x-y}{1-x+y} \right| + C$$

As curve passes through origin

$$0 = \frac{1}{2} \log \left| \frac{1+0-0}{1-0+0} \right| + C$$

$$\Rightarrow C = 0$$

$$x = \frac{1}{2} \log \left| \frac{1+x-y}{1-x+y} \right|$$

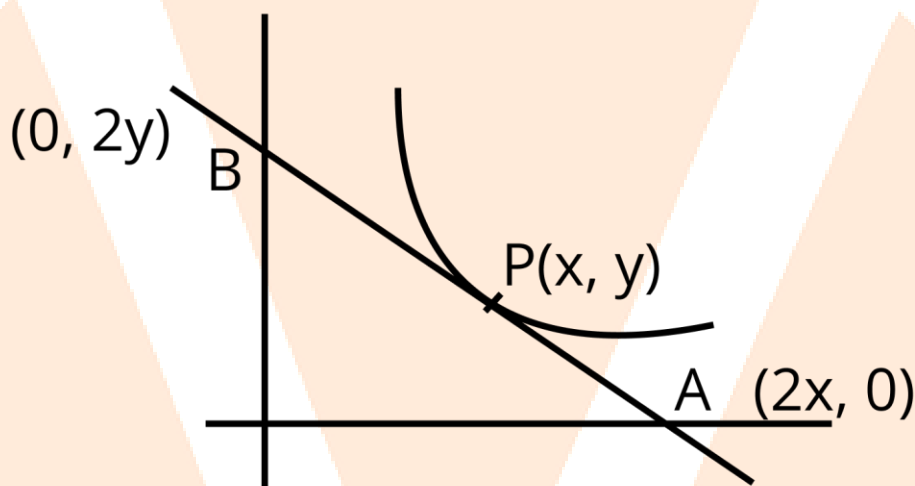
$$\Rightarrow 2x = \log \left| \frac{1+x-y}{1-x+y} \right|$$

$$e^{2x} = \left| \frac{1+x-y}{1-x+y} \right|$$

$$\Rightarrow (1-x+y)e^{2x} = 1+x-y$$

32. Find the equation of a curve passing through the point (1,1) . If the tangent drawn at any point P(x,y) on the curve meets the co-ordinate axes at A and B such that P is the mid-point of AB

Ans: The below figure obtained by the given information



Let the coordinate of the point P is (x,y) . It is given that, P is mid-point of AB

So, the coordinates of points A and B are (2x,0) , (0,2y) , respectively.

$$\text{Slope of AB} = \frac{0-2y}{2x-0} = -\frac{y}{x}$$

Since, the segment AB is a tangent to the curve at P

$$\frac{dy}{dx} = -\frac{y}{x}$$

$$\Rightarrow \frac{dy}{y} = -\frac{dx}{x}$$

Integrate both sides we get:

$$\log y = -\log x + \log C$$

$$\Rightarrow \log y = \log \frac{C}{x}$$

As given curve passes through (1,1)

$$\log 1 = \log \frac{C}{1}$$

$$\Rightarrow 0 = \log C$$

$$\Rightarrow c = 1$$

$$\log y = \log \frac{1}{x}$$

$$\Rightarrow y = \frac{1}{x}$$

$$\Rightarrow xy = 1$$

33. Solve $x \frac{dy}{dx} = y(\log y - \log x + 1)$

Ans: We have $x \frac{dy}{dx} = y(\log y - \log x + 1)$

$$x \frac{dy}{dx} = y \log \left(\frac{y}{x} + 1 \right) \quad x \frac{dy}{dx} = y \left(\log \frac{y}{x} + 1 \right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x} \left(\log \frac{y}{x} + 1 \right)$$

which is a homogeneous equation

Put $\frac{y}{x} = v$ or $y = vx$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

Substitute values we have

$$v+x \frac{dv}{dx} = v(\log v+1)$$

$$\Rightarrow x \frac{dv}{dx} = v(\log v+1-1)$$

$$\Rightarrow x \frac{dv}{dx} = v(\log v)$$

$$\Rightarrow \frac{dv}{v \log v} = \frac{dx}{x}$$

Integrate both sides

$$\int \frac{dv}{v \log v} = \int \frac{dx}{x}$$

Put $\log v = u$ in LHS we get

$$\frac{1}{v} \times dv = du$$

$$\Rightarrow \int \frac{du}{u} = \int \frac{dx}{x}$$

$$\log u = \log x + \log C$$

$$\Rightarrow \log u = \log Cx$$

$$\Rightarrow u = Cx$$

$$\log v = Cx$$

$$\Rightarrow \log \left(\frac{y}{x} \right) = Cx$$

Objective Questions

Choose the correct answer from the given four options in each of the Exercises from 34 to 75 (M.C.Q)

34. The degree of the differential equation $\left(\frac{d^2 y}{dx^2} \right)^2 + \left(\frac{dy}{dx} \right)^2 = x \sin \left(\frac{dy}{dx} \right)$ is:

(A) 1

(B) 2

(C) 3

(D) Not defined

Ans: Correct option (D) as degree of the differential equation is not defined because when we expand $\sin\left(\frac{dy}{dx}\right)$ we get an infinite series in the increasing power of $\frac{dy}{dx}$. Therefore, its degree is not defined.

35. The degree of the differential equation $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}} = \frac{d^2y}{dx^2}$ is

(A) 4

(B) $\frac{3}{2}$

(C) Not defined

(D) 2

Ans: Correct option (D)

As given that $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{\frac{3}{2}} = \frac{d^2y}{dx^2}$

Squaring both sides:

$$\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = \left(\frac{d^2y}{dx^2}\right)^2$$

Thus, degree is two

36. The order and degree of the differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^{\frac{1}{4}} + x^{\frac{1}{3}} = 0$, respectively, are

(A) 2 and not defined

(B) 2, 2

(C) 2, 3

(D) 3, 3

Ans: Correct option (A) Given that $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^{\frac{1}{4}} = -x^{\frac{1}{3}}$

$$\left(\frac{dy}{dx}\right)^{\frac{1}{4}} = -\left(x^{\frac{1}{5}} + \frac{d^2y}{dx^2}\right)$$

Squaring both sides

$$\left(\frac{dy}{dx}\right)^{\frac{1}{2}} = \left(x^{\frac{1}{5}} + \frac{d^2y}{dx^2}\right)^2$$

Again squaring both sides

$$\frac{dy}{dx} = \left(x^{\frac{1}{5}} + \frac{d^2y}{dx^2}\right)^4$$

Order is two, degree does not defined when we expand $\left(x^{\frac{1}{5}} + \frac{d^2y}{dx^2}\right)^4$, it cannot be in polynomial expansion.

37. If $y = e^x(A\cos x + B\sin x)$ then y is a solution of

(a) $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} = 0$

(b) $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$

$$(c) \frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = 0$$

$$(d) \frac{d^2y}{dx^2} + 2y = 0$$

Ans: Correct option (c)

As we have $y = e^{-x}(A\cos x + B\sin x)$

Differentiate both sides

$$\frac{dy}{dx} = -e^{-x}(A\cos x + B\sin x) + e^{-x}(-A\sin x + B\cos x)$$

$$\frac{dy}{dx} = -y + e^{-x}(-A\sin x + B\cos x)$$

Again differentiate both sides

$$\frac{d^2y}{dx^2} = \frac{-dy}{dx} + e^{-x}(-\cos x - B\sin x) - e^{-x}(-A\sin x + B\cos x)$$

$$\frac{d^2y}{dx^2} = \frac{-dy}{dx} - y - \left[\frac{dy}{dx} + y \right]$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\frac{dy}{dx} - y - \frac{dy}{dx} - y$$

$$\Rightarrow \frac{d^2y}{dx^2} = -2\frac{dy}{dx} - 2y$$

$$\Rightarrow \frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 2y = 0$$

38. The differential equation for $y = A\cos ax + B\sin ax$ where A and B are arbitrary constants is

$$(a) \frac{d^2y}{dx^2} - a^2y = 0$$

(b) $\frac{d^2y}{dx^2} + \alpha^2 y = 0$

(c) $\frac{d^2y}{dx^2} + \alpha y = 0$

(d) $\frac{d^2y}{dx^2} - \alpha y = 0$

Ans: Correct option (b)

As we have $y = A\cos\alpha x + B\sin\alpha x$

$$\frac{dy}{dx} = -\alpha A\sin\alpha x + \alpha B\cos\alpha x$$

Differentiate both sides:

$$\frac{d^2y}{dx^2} = -A\alpha^2\cos\alpha x - \alpha^2 B\sin\alpha x$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\alpha^2 (A\cos\alpha x + B\sin\alpha x)$$

$$\Rightarrow \frac{d^2y}{dx^2} = -\alpha^2 y$$

$$\Rightarrow \frac{d^2y}{dx^2} + \alpha^2 y = 0$$

39. Solution of differential equation $x dy - y dx = 0$ represents:

(A) a rectangular hyperbola

(B) parabola whose vertex is at origin

(C) straight line passing through origin

(D) a circle whose centre is at origin

Ans: Correct option (c)

As we have $x dy - y dx = 0$

$$x dy = y dx$$

$$\Rightarrow \frac{dy}{y} = \frac{dx}{x}$$

Integrate both sides:

$$\int \frac{dy}{y} = \int \frac{dx}{x}$$

$$\Rightarrow \log y = \log x + \log C$$

$$\Rightarrow \log y = \log Cx$$

$$\Rightarrow y = Cx$$

which is a straight line passing through origin

40. Integrating factor of the differential equation $\cos x \frac{dy}{dx} + y \sin x = 1$ is

(A) $\cos x$

(B) $\tan x$

(C) $\sec x$

(D) $\sin x$

Ans: Correct option (c)

As we have $\cos x \frac{dy}{dx} + y \sin x = 1$

Divide both side by $\cos x$ we get:

$$\frac{dy}{dx} + y \tan x = \sec x$$

Here $P = \tan x$, $Q = \sec x$

$$IF = e^{\int P dx} = e^{\int \tan x dx} = e^{\log \sec x} = \sec x$$

41. Solution of the differential equation $\tan y \sec^2 x dx + \tan x \sec^2 y dy = 0$ is

(a) $\tan x + \tan y = k$

(b) $\tan x - \tan y = k$

(c) $\frac{\tan x}{\tan y} = k$

(d) $\tan x \tan y = k$

Ans: Correct option (d)

We have $\tan y \sec^2 x dx + \tan x \sec^2 y dy = 0$

$$\tan y \sec^2 x dx = -\tan x \sec^2 y dy$$

$$\Rightarrow \frac{\sec^2 x}{\tan x} dx = \frac{-\sec^2 y}{\tan y} dy$$

Integrate both sides:

$$\int \frac{\sec^2 x}{\tan x} dx = -\int \frac{\sec^2 y}{\tan y} dy$$

Put $\tan x = t$ in LHS we get

$$\sec^2 x dx = dt$$

Put $\tan y = u$ in RHS we get

$$\sec^2 y dy = du$$

Substitute values we get:

$$\int \frac{dt}{t} = -\int \frac{du}{u}$$

$$\Rightarrow \log t = -\log u + \log k$$

$$\Rightarrow \log(t.u) = \log k$$

$$\Rightarrow \log(\tan x \tan y) = \log k$$

$$\Rightarrow \tan x \tan y = k$$

42. Family $y = Ax + A^3$ of curves is represented by the differential equation of degree

- (a) 1
- (b) 2
- (c) 3
- (d) 4

Ans: Correct option (a)

We have $y = Ax + A^3$

$\frac{dy}{dx} = A$ with degree one

43. Integrating factor of $\frac{x dy}{dx} - y = x^4 - 3x$ is

- (a) x
- (b) $\log x$
- (c) $\frac{1}{x}$
- (d) -x

Ans: Correct option (c)

We have $\frac{x dy}{dx} - y = x^4 - 3x$

Divide both sides by x we get

$$\frac{dy}{dx} - \frac{y}{x} = x^3 - 3$$

Here $P = -\frac{1}{x}$, $Q = x^3 - 3$

$$IF = e^{\int P dx} = e^{-\int \frac{1}{x} dx} = e^{-\log x} = \frac{1}{x}$$

44. Solution of $\frac{dy}{dx} - y = 1, y(0) = 1$ is given by

- (a) $xy = -e^x$
- (b) $xy = -e^{-x}$
- (c) $xy = -1$
- (d) $y = 2e^x - 1$

Ans: Correct option (b)

We have $\frac{dy}{dx} - y = 1$

$$\frac{dy}{dx} = 1+y$$

$$\Rightarrow \frac{dy}{1+y} = dx$$

Integrate both sides:

$$\int \frac{dy}{1+y} = \int dx$$

$$\Rightarrow \log(1+y) = x+C$$

When $x = 0, y = -1$ then

$$\log 2 = 0+C$$

$$\Rightarrow C = \log 2$$

Required solution

$$\log(1+y) = x + \log 2$$

$$\Rightarrow \log\left(\frac{1+y}{2}\right) = x$$

$$\Rightarrow \frac{1+y}{2} = e^x$$

$$\Rightarrow 1+y = 2e^x$$

$$\Rightarrow y = 2e^x - 1$$

45. The number of solutions of $\frac{dy}{dx} = \frac{y+1}{x-1}$, $y(1) = 2$ is

(a) none

(b) one

(c) two

(d) infinite

Ans: Correct option (b)

As we have $\frac{dy}{dx} = \frac{y+1}{x-1}$

$$\frac{dy}{y+1} = \frac{dx}{x-1}$$

Integrate both sides

$$\int \frac{dy}{y+1} = \int \frac{dx}{x-1}$$

$$\Rightarrow \log(y+1) = \log(x-1) - \log C$$

$$\Rightarrow C(y+1) = (x-1)$$

$$\Rightarrow C = \frac{x-1}{y+1}$$

When $x = 1$, $y = 2 \Rightarrow C = 0$

The required solution is $x-1 = 0$

Thus, one solution exists.

46. Which of the following is a second order differential equation

(a) $(y')^2 + x = y^2$

(b) $y' y'' + y = \sin x$

(c) $y''' + (y'')^2 + y = 0$

(d) $y' = y^2$

Ans: Correct option (b)

47. Integrating factor of the differential equation $(1-x^2)\frac{dy}{dx} - xy = 1$ is

(a) $-x$

(b) $\frac{x}{1+x^2}$

(c) $\sqrt{1-x^2}$

(d) $\frac{1}{2} \log(1-x^2)$

Ans: Correct option (c)

We have $(1-x^2)\frac{dy}{dx} - xy = 1$

Divide both sides by $(1-x^2)$ we get

$\frac{dy}{dx} - \frac{x}{1-x^2}y = \frac{1}{1-x^2}$ is a linear differential equation

$IF = e^{-\int \frac{x}{1-x^2} dx}$

Put $1-x^2 = t$

$$-2x \, dx = dt$$

$$\Rightarrow x \, dx = -\frac{dt}{2}$$

$$\text{Now, IF} = e^{\frac{1}{2} \int \frac{dt}{t}} = e^{\frac{1}{2} \log t} = e^{\frac{1}{2} \log(1-x^2)} = \sqrt{1-x^2}$$

48. $\tan^{-1}x + \tan^{-1}y = c$ is the general solution of the differential equation

(a) $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$

(b) $\frac{dy}{dx} = \frac{1+x^2}{1+y^2}$

(c) $(1+x^2)dy + (1+y^2)dx = 0$

(d) $(1+x^2)dx + (1+y^2)dy = 0$

Ans: Correct option (c)

We have $\tan^{-1}x + \tan^{-1}y = c$

Differentiate both sides we get:

$$\frac{d}{dx}(\tan^{-1}x + \tan^{-1}y) = \frac{d}{dx}c$$

$$\Rightarrow \frac{1}{1+x^2} + \frac{1}{1+y^2} \times \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{1}{1+x^2} + \frac{1}{1+y^2} \times \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{1}{1+y^2} \times \frac{dy}{dx} = -\frac{1}{1+x^2}$$

$$\Rightarrow (1+x^2)dy + (1+y^2)dx = 0$$

49. The differential equation $y \frac{dy}{dx} + x = c$ represents:

- (A) Family of hyperbolas
- (B) Family of parabolas
- (C) Family of ellipses
- (D) Family of circles

Ans: Correct option (D) Given that $y \frac{dy}{dx} + x = c$

$$y \frac{dy}{dx} = C - x$$

$$\Rightarrow y \, dy = (C - x) \, dx$$

Integrate both sides

$$\int y \, dy = \int (C - x) \, dx$$

$$\Rightarrow \frac{y^2}{2} = Cx - \frac{x^2}{2} + k$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} = Cx + K$$

$$\Rightarrow \frac{x^2}{2} + \frac{y^2}{2} - Cx = k$$

represents family of circles

50. The general solution of $e^2 \cos y \, dx - e^x \sin y \, dy = 0$ is

- (a) $e^x \cos y = k$
- (b) $e^x \sin y = k$
- (c) $e^x = k \cos y$
- (d) $e^x = k \sin y$

Ans: Correct option (a)

We have $e^x \cos y dx - e^x \sin y dy = 0$

$$e^x \cos y dx = e^x \sin y dy = 0$$

$$\Rightarrow e^x \cos y dx = e^x \sin y dy$$

$$\Rightarrow \frac{dx}{dy} = \tan y$$

$$\Rightarrow dx = \tan y dy$$

Integrate both sides

$$\int dx = \int \tan y dy$$

$$\Rightarrow x = \log \sec y + C$$

$$\Rightarrow x - C = \log \sec y$$

$$\Rightarrow \sec y = e^{x-C}$$

$$\Rightarrow \sec y = e^x e^{-C}$$

$$\Rightarrow \frac{1}{\cos y} = \frac{e^x}{e^C}$$

$$\Rightarrow e^x \cos y = e^C$$

$$\Rightarrow e^x \cos y = K \quad \left[\text{where } K = e^C \right]$$

51. The degree of the differential equation $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 + 6y^5 = 0$ is

(a) 1

(b) 2

(c) 3

(d) 5

Ans: Correct option (a) we have $\frac{d^2y}{dx^2} + \left(\frac{dy}{dx}\right)^3 + 6y^5 = 0$

As we know degree of differential equation is exponent of highest order derivative

So, degree is 1

52. The solution of $\frac{dy}{dx} + y = e^{-x}$, $y(0) = 0$

(a) $y = e^x(x-1)$

(b) $y = xe^{-x}$

(c) $y = xe^{-x} + 1$

(d) $y = (x+1)e^{-x}$

Ans: Correct option (b)

We have $\frac{dy}{dx} + y = e^{-x}$

Here $P = 1$, $Q = e^{-x}$

IF $= e^{\int P dx} = e^{\int dx} = e^x$

General solution is

$$y \times e^x = \int e^{-x} e^x dx + C$$

$$\Rightarrow y \times e^x = \int dx + C$$

$$\Rightarrow y \times e^x = x + C$$

When $x = 0$, $y = 0 \Rightarrow 0 = C$

We have

$$y \times e^x = x$$

$$\Rightarrow y = xe^{-x}$$

53. Integrating factor of the differential equation $\frac{dy}{dx} + y \tan x - \sec x = 0$

(a) $\cos x$

(b) $\sec x$

(c) $e^{\cos x}$

(d) $e^{\sec x}$

Ans: Correct option (b) we have $\frac{dy}{dx} + y \tan x - \sec x = 0$

$$\frac{dy}{dx} + y \tan x = \sec x$$

Here $P = \tan x$, $Q = \sec x$

$$IF = e^{\int P dx} = e^{(\log \sec x)} = \sec x$$

54. The Solution of the differential equation $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$ is

(a) $y = \tan^{-1}x$

(b) $y - x = k(1 + xy)$

(c) $x = \tan^{-1}y$

(d) $\tan(xy) = k$

Ans: Correct option (b) we have $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$

$$\frac{dy}{1+y^2} = \frac{dx}{1+x^2}$$

Integrate both sides we get:

$$\int \frac{dy}{1+y^2} = \int \frac{dx}{1+x^2}$$

$$\Rightarrow \tan^{-1}y = \tan^{-1}x + C$$

$$\Rightarrow \tan^{-1}y - \tan^{-1}x = C$$

$$\Rightarrow \tan^{-1}\left(\frac{y-x}{1+xy}\right) = C$$

$$\Rightarrow \frac{y-x}{1+xy} = \tan C$$

$$\Rightarrow y-x = \tan C(1+xy)$$

$$\Rightarrow y-x = K(1+xy)$$

where $k = \tan C$

55. The integrating factor of the differential equation $\frac{dy}{dx} + y = \frac{1+y}{x}$ is

(a) $\frac{x}{e^x}$

(b) $\frac{e^x}{x}$

(c) xe^x

(d) e^x

Ans: Correct option (b) we have $\frac{dy}{dx} + y = \frac{1+y}{x}$

$$\frac{dy}{dx} = \frac{1+y}{x} - y$$

$$\Rightarrow \frac{dy}{dx} = \frac{1+y-xy}{x}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{x} + \frac{y(1-x)}{x}$$

$$\Rightarrow \frac{dy}{dx} - \left(\frac{1-x}{x}\right)y = \frac{1}{x}$$

Here $P = \frac{-(1-x)}{x}$, $Q = \frac{1}{x}$

$$IF = e^{\int P dx} = e^{-\int \frac{1-x}{x} dx} = e^{\int \frac{x-1}{x} dx} = e^{\int \left(1 - \frac{1}{x}\right) dx} = e^x \times e^{\log\left(\frac{1}{x}\right)} = e^x \times \frac{1}{x}$$

56. $y = ae^{mx} + be^{-mx}$ satisfies which of the following differential equation

(a) $\frac{dy}{dx} + my = 0$

(b) $\frac{dy}{dx} - my = 0$

(c) $\frac{d^2y}{dx^2} - m^2y = 0$

(d) $\frac{d^2y}{dx^2} + m^2y = 0$

Ans: Correct option (c) we have $y = ae^{mx} + be^{-mx}$

Differentiate both sides we get:

$$\frac{dy}{dx} = mae^{mx} - bme^{-mx}$$

Again differentiate both sides

$$\Rightarrow \frac{d^2y}{dx^2} = m^2ae^{mx} + bm^2e^{-mx}$$

$$\Rightarrow \frac{d^2y}{dx^2} = m^2(ae^{mx} + be^{-mx})$$

$$\Rightarrow \frac{d^2y}{dx^2} = m^2y$$

$$\Rightarrow \frac{d^2y}{dx^2} - m^2y = 0$$

57. The solution of the differential equation $\cos x \sin y \, dx + \sin x \cos y \, dy = 0$ is

(a) $\frac{\sin x}{\sin y} = C$

(b) $\sin x \sin y = C$

(c) $\sin x + \sin y = C$

(d) $\cos x \cos y = C$

Ans: Correct option (b) we have $\cos x \sin y dx + \sin x \cos y dy = 0$

$$\cos x \sin y dx = -\sin x \cos y dy$$

$$\Rightarrow \frac{\cos x}{\sin x} dx = -\frac{\cos y}{\sin y} dy$$

$$\Rightarrow \cot x dx = -\cot y dy$$

Integrate both sides

$$\int \cot x dx = \int -\cot y dy$$

$$\Rightarrow \log \sin x = -\log \sin y + \log C$$

$$\Rightarrow \log \sin x \sin y = \log C$$

$$\Rightarrow \sin x \sin y = C$$

58. The solution of $x \frac{dy}{dx} + y = e^x$ is

(a) $y = \frac{e^x}{x} + \frac{k}{x}$

(b) $y = xe^x + cx$

(c) $y = xe^x + k$

(d) $x = \frac{e^y}{y} + \frac{k}{y}$

Ans: Correct option (a) we have $x \frac{dy}{dx} + y = e^x$

Divide both sides by x we get:

$$\frac{dy}{dx} + \frac{y}{x} = \frac{e^x}{x} \text{ is a linear differential equation}$$

$$IF = e^{\int \frac{1}{x} dx} = e^{(\log x)} = x$$

$$\text{General solution is } y \cdot x = \int \left(\frac{e^x}{x} \cdot x \right) dx \cdot x = \int \left(\frac{e^x}{x} \cdot x \right) dx$$

$$\Rightarrow y \times x = \int e^x dx$$

$$\Rightarrow y \times x = e^x + k$$

$$\Rightarrow y = \frac{e^x}{x} + \frac{k}{x}$$

59. The differential equation of the family of curves $x^2 + y^2 - 2ay = 0$ where a is arbitrary constant is

(a) $(x^2 - y^2) \frac{dy}{dx} = 2xy$

(b) $2(x^2 + y^2) \frac{dy}{dx} = xy$

(c) $2(x^2 - y^2) \frac{dy}{dx} = xy$

(d) $(x^2 + y^2) \frac{dy}{dx} = 2xy$

Ans: Correct option (a) we have $x^2 + y^2 - 2ay = 0$

$$\frac{x^2 + y^2}{y} = 2a$$

Differentiate both sides, we get:

$$\frac{y \left(2x + 2y \frac{dy}{dx} \right) - (x^2 + y^2) \frac{dy}{dx}}{y^2} = 0$$

$$\Rightarrow 2xy + 2y^2 \frac{dy}{dx} - (x^2 + y^2) \frac{dy}{dx} = 0$$

$$\Rightarrow (2y^2 - x^2 - y^2) \frac{dy}{dx} = -2xy$$

$$\Rightarrow (y^2 - x^2) \frac{dy}{dx} = -2xy$$

$$\Rightarrow (x^2 - y^2) \frac{dy}{dx} = 2xy$$

60. Family $y = Ax + A^3$ of curves will correspond to a differential equation of order

- (a) 3
- (b) 2
- (c) 1
- (d) not defined

Ans: Correct option (c) we have $y = Ax + A^3$

$$\frac{dy}{dx} = A$$

Replace A by $\frac{dy}{dx}$ we have

$$y = x \frac{dy}{dx} + \left(\frac{dy}{dx} \right)^3$$

So, order is 1

61. The general solution is $\frac{dy}{dx} = 2xe^{x^2-y}$

- (a) $e^{x^2-y} = c$
- (b) $e^{-y} + e^{x^2} = c$
- (c) $e^y = e^{x^2} + c$
- (d) $e^{x^2+y} = c$

Ans: Correct option (c) we have $\frac{dy}{dx} = 2xe^{x^2-y} = 2xe^{x^2} \times e^{-y}$

$$e^y \frac{dy}{dx} = 2xe^{x^2}$$

$$\Rightarrow e^y dy = 2xe^{x^2} dx$$

Integrate both sides:

$$\int e^y dy = 2 \int xe^{x^2} dx$$

Put $x^2 = t$ in RHS we get

$$2x dx = dt$$

$$\Rightarrow \int e^y dy = \int e^t dt$$

$$\Rightarrow e^y = e^t + C$$

$$\Rightarrow e^y = e^{x^2} + C$$

62. The curve for which the slope of the tangent at any point is equal to the ratio of the abscissa to the ordinate of the point is:

(A) an ellipse

(B) parabola

(C) circle

(D) rectangular hyperbola

Ans: Correct option (D) Slope of tangent to the curve = $\frac{dy}{dx}$

Ratio of abscissa to ordinate = $\frac{x}{y}$

Accordingly, $\frac{dy}{dx} = \frac{x}{y}$

$$y dy = x dx$$

Integrate both sides:

$$\int y \, dy = \int x \, dx$$

$$\Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + C$$

$$\Rightarrow \frac{y^2}{2} - \frac{x^2}{2} = C$$

$y^2 - x^2 = 2C$ is an equation of rectangular hyperbola

63. The general solution of the differential equation $\frac{dy}{dx} e^{\frac{x^2}{2}} + xy \frac{dy}{dx} = e^{\frac{x^2}{2}} + xy$

(a) $y = ce^{\frac{-x^2}{2}}$

(b) $y = ce^{\frac{x^2}{2}}$

(c) $y = (x + c)e^{\frac{x^2}{2}}$

(d) $y = (c - x)e^{\frac{x^2}{2}}$

Ans: Correct option (c) we have $\frac{dy}{dx} e^{\frac{x^2}{2}} + xy \frac{dy}{dx} = e^{\frac{x^2}{2}} + xy$

$\frac{dy}{dx} - xy = e^{\frac{x^2}{2}}$ is a linear differential equation

Here $P = -x$, $Q = e^{\frac{x^2}{2}}$

$$IF = e^{\int -x \, dx} = e^{\frac{-x^2}{2}}$$

General solution

$$ye^{\frac{-x^2}{2}} = \int e^{\frac{-x^2}{2}} \cdot e^{\frac{x^2}{2}} dx + C ye^{\frac{-x^2}{2}} = \int e^{\frac{-x^2}{2}} \times e^{\frac{x^2}{2}} dx + C$$

$$\Rightarrow ye^{\frac{-x^2}{2}} = \int 1 dx + C$$

$$\Rightarrow y \times e^{\frac{-x^2}{2}} = x + C$$

$$\Rightarrow y = xe^{\frac{x^2}{2}} + Ce^{\frac{-x^2}{2}}$$

$$\Rightarrow y = (x + C)e^{\frac{x^2}{2}}$$

64. The solution of the equation $(2y - 1)dx - (2x + 3)dy = 0$ is

(a) $\frac{2x-1}{2y+3} = k$

(b) $\frac{2y+1}{2x-3} = k$

(c) $\frac{2x+3}{2y-1} = k$

(d) $\frac{2x-1}{2y-1} = k$

Ans: Correct option (c) we have $(2y-1)dx - (2x+3)dy = 0$

$$(2y-1)dx = (2x+3)dy$$

$$\Rightarrow \frac{dx}{2x+3} = \frac{dy}{2y-1}$$

Integrate both sides:

$$\begin{aligned}\int \frac{dx}{2x+3} &= \int \frac{dy}{2y-1} \\ \Rightarrow \frac{1}{2} \log(2x+3) &= \frac{1}{2} \log(2y-1) + \log C \\ \Rightarrow \frac{1}{2} [\log(2x+3) - \log(2y-1)] &= \log C \\ \Rightarrow \frac{1}{2} \log \left(\frac{2x+3}{2y-1} \right) &= \log C \\ \Rightarrow \left(\frac{2x+3}{2y-1} \right)^{\frac{1}{2}} &= C \\ \Rightarrow \frac{2x+3}{2y-1} &= C^2 \\ \Rightarrow \frac{2x+3}{2y-1} &= k \quad \text{where } K = C^2\end{aligned}$$

65. The differential equation for which $y = a \cos x + b \sin x$ is a solution, is

- (a) $\frac{d^2y}{dx^2} + y = 0$
- (b) $\frac{d^2y}{dx^2} - y = 0$
- (c) $\frac{d^2y}{dx^2} + (a+b)y = 0$
- (d) $\frac{d^2y}{dx^2} + (a-b)y = 0$

Ans: Correct option (a) we have $y = a \cos x + b \sin x$

Differentiate both sides we get:

$$\frac{dy}{dx} = -a \sin x + b \cos x$$

Again, differentiate both sides:

$$\frac{d^2y}{dx^2} = -a \sin x + b \cos x$$

$$\Rightarrow \frac{d^2y}{dx^2} = -a \cos x - b \sin x$$

$$\Rightarrow \frac{d^2y}{dx^2} = -(a \cos x + b \sin x)$$

$$\Rightarrow \frac{d^2y}{dx^2} = -y$$

$$\Rightarrow \frac{d^2y}{dx^2} + y = 0$$

66. The solution of $\frac{dy}{dx} + y = e^{-x}$, $y(0) = 0$ is

(a) $y = e^{-x}(x-1)$

(b) $y = xe^x$

(c) $y = xe^{-x} + 1$

(d) $y = xe^{-x}$

Ans: Correct option (d) we have $\frac{dy}{dx} + y = e^{-x}$ is a linear differential equation

Here $P = 1$, $Q = e^{-x}$

$$IF = e^x$$

General solution is

$$y \times e^x = \int e^{-x} \times e^x dx + C$$

$$\Rightarrow ye^x = \int dx + C$$

$$\Rightarrow ye^x = x + C$$

When $x = 0$, $y = 0$ then $0 = C$

Then, we have

$$y.e^x = x$$

$$\Rightarrow y = xe^{-x}$$

67. The order and degree of the differential equation $\left(\frac{d^3y}{dx^3}\right)^2 - 3\frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^4 = y^4$

(a) 1,4

(b) 3,4

(c) 2,4

(d) 3,2

Ans: Correct option (d) we have $\left(\frac{d^3y}{dx^3}\right)^2 - 3\frac{d^2y}{dx^2} + 2\left(\frac{dy}{dx}\right)^4 = y^4$

Order is 3 and degree is 2

68. The order and degree of the differential equation $\left[1 + \left(\frac{dy}{dx}\right)^2\right] = \frac{d^2y}{dx^2}$ are

(a) $2, \frac{3}{2}$

(b) 2,3

(c) 2,1

(d) 3,4

Ans: Correct option (c) we have $\left[1 + \left(\frac{dy}{dx}\right)^2\right] = \frac{d^2y}{dx^2}$

Order is 2 and degree is 1

69. The differential equation of the family of curves $y^2 = 4a(x + a)$ is

(a) $y^2 = 4 \frac{dy}{dx} \left(x + \frac{dy}{dx} \right)$

(b) $2y \frac{dy}{dx} = 4a$

(c) $y \frac{d^2y}{dx^2} + \left(\frac{dy}{dx} \right)^2 = 0$

(d) $2x \frac{dy}{dx} + y \left(\frac{dy}{dx} \right)^2 - y = 0$

Ans: Correct option (d) we have $y^2 = 4a(x+a)$

Differentiate both sides we get:

$$2y \frac{dy}{dx} = 4a$$

$$\Rightarrow 2y \frac{dy}{dx} = 4a$$

$$\Rightarrow y \frac{dy}{dx} = 2a$$

$$\Rightarrow a = \frac{1}{2} y \frac{dy}{dx}$$

Now, put values we get

$$y^2 = 2y \frac{dy}{dx} = \left(x + \frac{1}{2} y \frac{dy}{dx} \right)$$

$$\Rightarrow y^2 = 2xy \frac{dy}{dx} + y^2 \left(\frac{dy}{dx} \right)^2$$

$$\Rightarrow 2x \frac{dy}{dx} + y \left(\frac{dy}{dx} \right)^2 - y = 0$$

70. Which of the following is the general solution of $\frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + y = 0$

(a) $y = (Ax + B)e^x$

(b) $y = (Ax + B)e^{-x}$

(c) $y = Ae^x + Be^{-x}$

(d) $y = A \cos x + B \sin x$

Ans: Correct option (a) we have $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 0$

$$D^2y - 2Dy + y = 0 \quad \text{where } D = \frac{d}{dx}$$

$$\Rightarrow (D^2 - 2D + 1)y = 0$$

Auxiliary equation is $m^2 - 2m + 1 = 0$

$$(m-1)^2 = 0$$

$$\Rightarrow m = 1, 1$$

As roots are real and equal

$$CF = (Ax+B)e^x \Rightarrow y = (Ax+B)e^x$$

71. General solution of $\frac{dy}{dx} + y \tan x = \sec x$

(a) $y \sec x = \tan x + c$

(b) $y \tan x = \sec x + c$

(c) $\tan x = y \tan x + c$

(d) $x \sec x = \tan y + c$

Ans: Correct answer (a) we have $\frac{dy}{dx} + y \tan x = \sec x$ is a linear differential equation

Here $P = \tan x$, $Q = \sec x$

$$IF = e^{\int \tan x \, dx} = e^{\log|\sec x|} = \sec x$$

General solution is

$$y \times \sec x = \int \sec x \times \sec x + C$$

$$\Rightarrow y \times \sec x = \int \sec^2 x \, dx + C$$

$$\Rightarrow y \times \sec x = \tan x + C$$

72. Solution of the differential equation $\frac{dy}{dx} + \frac{y}{x} = \sin x$

(a) $x(y + \cos x) = \sin x + c$

(b) $x(y - \cos x) = \sin x + c$

(c) $xy \cos x = \sin x + c$

(d) $x(y + \cos x) = \cos x + c$

Ans: Correct option (a) we have $\frac{dy}{dx} + \frac{y}{x} = \sin x$ is a linear differential equation

Here $P = \frac{1}{x}$, $Q = \sin x$

$$IF = e^{\int \frac{1}{x}} = e^{\log x} = x$$

General solution is

$$yx = \int x \sin x \, dx + C$$

Take $I = \int x \sin x \, dx$

$$= -x \cos x - \int -\cos x \, dx$$

$$= -x \cos x + \sin x$$

Put values we get

$$xy = -x \cos x + \sin x + C$$

$$\Rightarrow x(y + \cos x) = \sin x + C$$

73. The general solution of the differential equation $(e^x + 1)y \, dy = (y + 1)e^x \, dx$ is

(a) $(y + 1) = k(e^x + 1)$

(b) $y + 1 = e^x + 1 + k$

(c) $y = \log \{k(y + 1)(e^x + 1)\}$

(d) $y = \log \left\{ \frac{e^x + 1}{y + 1} \right\} + k$

Ans: Correct option (c) we have $(e^x + 1)y \, dy = (y + 1)e^x \, dx$

$$\frac{dy}{dx} = \frac{e^x(1+y)}{(e^x+1)y}$$

$$\Rightarrow \frac{dx}{dy} = \frac{(e^x+1)y}{e^x(1+y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{e^x y}{e^x(1+y)} + \frac{y}{e^x(1+y)}$$

$$\Rightarrow \frac{dx}{dy} = \frac{y}{1+y} + \frac{y}{(1+y)e^x}$$

$$\Rightarrow \frac{dx}{dy} = \frac{y}{1+y} \left(1 + \frac{1}{e^x} \right)$$

$$\Rightarrow \frac{dx}{dy} = \frac{y}{1+y} \left(\frac{e^x+1}{e^x} \right)$$

$$\Rightarrow \left(\frac{y}{1+y} \right) dy = \left(\frac{e^x}{e^x+1} \right) dx$$

Integrate both sides:

$$\int \frac{y}{1+y} dy = \int \frac{e^x}{1+e^x} dx$$

$$\int \frac{1+y-1}{1+y} dy = \int \frac{e^x}{1+e^x} dx$$

$$\Rightarrow \int 1 dy - \int \frac{1}{1+y} dy = \int \frac{e^x}{1+e^x} dx$$

$$\Rightarrow y = \log(1+y) + \log(1+e^x) + \log k$$

$$\Rightarrow y = \log\{k(1+y)(1+e^x)\}$$

74. The solution of the differential equation $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$ is

(a) $y = e^{x-y} - x^2 e^{-y} + c$

(b) $e^y - e^x = \frac{x^3}{3} + c$

(c) $e^x + e^y = \frac{x^3}{3} + c$

(d) $e^x - e^y = \frac{x^3}{3} + c$

Ans: Correct option (b) we have $\frac{dy}{dx} = e^{x-y} + x^2 e^{-y}$

$$\frac{dy}{dx} = e^x e^{-y} + x^2 e^{-y}$$

$$\Rightarrow \frac{dy}{dx} = \frac{e^x + x^2}{e^y}$$

$$\Rightarrow e^y dy = (e^x + x^2) dx$$

Integrate both sides:

$$\int e^y dy = \int (e^x + x^2) dx$$

$$\Rightarrow e^y = e^x + \frac{x^3}{3} + C$$

$$\Rightarrow e^y - e^x = \frac{x^3}{3} + C$$

75. The solution of the differential equation $\frac{dy}{dx} + \frac{2xy}{1+x^2} = \frac{1}{(1+x^2)^2}$ is

(a) $y(1+x^2) = c + \tan^{-1}x$

(b) $\frac{y}{1+x^2} = c + \tan^{-1}x$

(c) $y \log(1+x^2) = c + \tan^{-1}x$

(d) $y(1+x^2) = c + \sin^{-1}x$

Ans: Correct option (a) we have $\frac{dy}{dx} + \frac{2xy}{1+x^2} = \frac{1}{(1+x^2)^2}$

Here $P = \frac{2x}{1+x^2}$, $Q = \frac{1}{(1+x^2)^2}$ is a linear differential equation

$$IF = e^{\int \frac{2x}{1+x^2} dx}$$

Put $1+x^2 = t \Rightarrow 2x dx = dt$

So, $IF = e^{\int \frac{dt}{t}} = e^{\log t} = e^{\log(1+x^2)} = 1+x^2$

General solution is

$$y \times (1+x^2) = \int (1+x^2) \frac{1}{(1+x^2)^2} dx + C$$

$$\Rightarrow y(1+x^2) = \int \frac{1}{1+x^2} dx + C$$

$$\Rightarrow y(1+x^2) = \tan^{-1}x + C$$

76. Fill in the blanks of the following (i to xi)

(i) The degree of the differential equation $\frac{d^2y}{dx^2} + e^{\frac{dy}{dx}} = 0$ is _____

Ans: Given differential equation is $\frac{d^2y}{dx^2} + e^{\frac{dy}{dx}} = 0$

Degree of this equation is not defined.

(ii) The degree of the differential equation $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = x$ is _____

Ans: Given differential equation is $\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = x$

So, degree of this equation is two.

(iii) The number of arbitrary constants in the general solution of a differential equation of order three is _____

Ans: There are three arbitrary constants.

(iv) $\frac{dy}{dx} + \frac{y}{x \log x} = \frac{1}{x}$ is an equation of the type _____

Ans: given differential equation is $\frac{dy}{dx} + \frac{y}{x \log x} = \frac{1}{x}$

This is of type $\frac{dy}{dx} + Py = Q$

(v) General solution of the differential equation of the type $\frac{dy}{dx} + P_1x = Q_1$, $\frac{dx}{dy} + P_1x = Q_1$ is given by _____

Ans: Given differential equation is $\frac{dy}{dx} + P_1x = Q_1$, $\frac{dx}{dy} + P_1x = Q_1$

General solution is

$$x \cdot IF = \int Q(IF) dy + C$$

$$\Rightarrow x e^{\int P dy} = \int Q \left\{ e^{\int P dy} \right\} dy + C$$

$$\Rightarrow x \cdot e^{\int P_1 dy} = \int Q_1 \left\{ e^{\int P_1 dy} \right\} dy + C$$

(vi) The solution of differential equation $\frac{x dy}{dx} + 2y = x^2$ is _____

Ans: Given differential equation is $\frac{x dy}{dx} + 2y = x^2$ in the form $\frac{dy}{dx} + Py = Q$

$$IF = e^{\int \frac{2}{x} dx} = e^{2 \log x} = x^2$$

General solution is

$$yx^2 = \int xx^2 dx + C$$

$$\Rightarrow yx^2 = \frac{x^4}{4} + C$$

$$\Rightarrow y = \frac{x^2}{4} + Cx^2$$

(vii) The solution of $(1+x^2) \frac{dy}{dx} + 2xy - 4x^2 = 0$ is _____

Ans: Given differential equation is $(1+x^2) \frac{dy}{dx} + 2xy - 4x^2 = 0$

Divide both side by $(1+x^2)$

$$\frac{dy}{dx} + \frac{2xy}{1+x^2} - \frac{4x^2}{1+x^2} = 0$$

$$\Rightarrow \frac{dy}{dx} + \frac{2x}{1+x^2} y = \frac{4x^2}{1+x^2}$$

$$IF = e^{\int \frac{2x}{1+x^2} dx}$$

Put $1+x^2 = t$

$$2x dx = dt$$

$$IF = e^{\int \frac{dt}{2}} = e^{\log t} = e^{\log(1+x^2)} = 1+x^2$$

General solution is

$$y \times (1+x^2) = \int (1+x^2) \frac{4x^2}{(1+x^2)} dx + C$$

$$\Rightarrow (1+x^2)y = \int 4x^2 dx + C$$

$$\Rightarrow (1+x^2)y = 4 \frac{x^3}{3} + C$$

$$\Rightarrow y = \frac{4x^3}{3(1+x^2)} + C(1+x^2)^{-1}$$

(viii) The solution of the differential equation $y dx + (x + xy)dy = 0$

Ans: Given differential equation is $y dx + (x + xy)dy = 0$

$$y dx + x(1+y)dy = 0$$

$$\Rightarrow \frac{dx}{-x} = \left(\frac{1+y}{y} \right) dy$$

$$\Rightarrow \int \frac{1}{x} dx = - \int \left(\frac{1}{y} + 1 \right) dy$$

$$\Rightarrow \log(x) = -\log(y) - y + \log A$$

$$\Rightarrow \log(x) + \log(y) + y = \log A$$

$$\Rightarrow \log(xy) + y = \log A$$

$$\Rightarrow \log xy + \log e^y = \log A$$

$$\Rightarrow xy e^y = A$$

$$\Rightarrow xy = Ae^{-y}$$

(ix) General solution of $\frac{dy}{dx} + y = \sin x$ is _____

Ans: Given differential equation is $\frac{dy}{dx} + y = \sin x$

$$IF = \int e^{1 dx} = e^x$$

General solution is $y \times e^x = \int e^x \sin x dx + C$

Let $\int e^x \sin x \, dx$

$$\begin{aligned} I &= \sin x e^x - \int \cos x e^x \, dx \\ &= \sin x e^x - \cos x e^x + \int (-\sin x) e^x \, dx \\ 2I &= e^x (\sin x - \cos x) \\ I &= \frac{1}{2} e^x (\sin x - \cos x) \end{aligned}$$

Then we have

$$\begin{aligned} y \times e^x &= \frac{x}{2} (\sin x - \cos x) + C \\ \Rightarrow y &= \frac{1}{2} (\sin x - \cos x) + C \times e^{-x} \end{aligned}$$

(x) The solution of differential equation $\cot y \, dx = x \, dy$ is _____

Ans: Given differential equation is $\cot y \, dx = x \, dy$

$$\frac{1}{x} \, dx = \tan y \, dy$$

Integrate both sides we get:

$$\begin{aligned} \int \frac{1}{x} \, dx &= \int \tan y \, dy \\ \Rightarrow \log(x) &= \log(\sec y) + \log C \\ \Rightarrow \log\left(\frac{x}{\sec y}\right) &= \log C \\ \Rightarrow \frac{x}{\sec y} &= C \\ \Rightarrow x &= C \sec y \end{aligned}$$

(xi) The integrating factor of $\frac{dy}{dx} + y = \frac{1+y}{x}$ is _____

Ans: Given differential equation is $\frac{dy}{dx} + y = \frac{1+y}{x}$

$$\frac{dy}{dx} + y = \frac{1}{x} + \frac{y}{x}$$

$$\Rightarrow \frac{dy}{dx} + y \left(1 - \frac{1}{x}\right) = \frac{1}{x}$$

$$\text{IF} = e^{\int \left(1 - \frac{1}{x}\right) dx} = e^{x - \log x} = e^x e^{-\log x} = \frac{e^x}{x}$$

77. State true or false of the following

(i) Integrating factor of the differential equation of the form $\frac{dy}{dx} + p_1x = Q_1$ is given by $e^{\int p_1 dy}$

Ans: True

Given differential equation $\frac{dy}{dx} + p_1x = Q_1$

IF is equal to $e^{\int p_1 dy}$

(ii) Solution of the differential equation of the type $\frac{dy}{dx} + p_1x = Q_1$ is given by $x \text{ I.F.} = \int (\text{I.F.}) \times Q_1 dy$

Ans: True

(iii) Correct substitution for the solution of the differential equation of the type $\frac{dy}{dx} = f(x, y)$ where $f(x, y)$ is a homogeneous function of zero degree is $y = vx$

Ans: True

(iv) Correct substitution for the solution of the differential equation of the type $\frac{dy}{dx} = g(x, y)$ where $g(x, y)$ is a homogeneous function of the degree zero is $x = vy$

Ans: True

(v) Number of arbitrary constants in the particular solution of a differential equation of order two is two.

Ans: False, There is no arbitrary constant in the particular solution of a differential equation.

(vi) The differential equation representing the family of circles $x^2 + (y - a)^2 = a^2$ will be of order two.

Ans: False, As We know order of differential equation is equal to number of arbitrary constant.

Here, number of arbitrary constant is equal to one. Thus, order is one.

(vii) The solution of $\frac{dy}{dx} = \left(\frac{y}{x}\right)^{\frac{1}{3}}$ is $y^{\frac{2}{3}} - x^{\frac{2}{3}} = c$

Ans: True, given differential equation $\frac{dy}{dx} = \left(\frac{y}{x}\right)^{\frac{1}{3}}$

$$\frac{dy}{dx} = \frac{y^{\frac{1}{3}}}{x^{\frac{1}{3}}}$$

$$\Rightarrow y^{\frac{-1}{3}} dy = x^{\frac{-1}{3}} dx$$

Integrate both sides:

$$\int y^{\frac{-1}{3}} dy = \int x^{\frac{-1}{3}} dx$$

$$\Rightarrow \frac{y^{\frac{-1}{3}+1}}{\frac{-1}{3}+1} = \frac{x^{\frac{-1}{3}+1}}{\frac{-1}{3}+1} + C'$$

$$\Rightarrow \frac{3}{2} y^{\frac{2}{3}} = \frac{3}{2} x^{\frac{2}{3}} + C'$$

$$\Rightarrow y^{\frac{2}{3}} - x^{\frac{2}{3}} = C'$$

(viii) Differential equation representing the family of curve

$y = e^x (A \cos x + B \sin x)$ is $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = 0$

Ans: True,

Given that $y = e^x (A \cos x + B \sin x)$

Differentiate both sides we get:

$$\frac{dy}{dx} = e^x (-A \sin x + B \cos x) + e^x (A \cos x + B \sin x)$$

$$\frac{dy}{dx} - y = e^x (-A \sin x + B \cos x)$$

Again, differentiate both sides:

$$\frac{d^2y}{dx^2} - \frac{dy}{dx} = e^x (-A \cos x - B \sin x) + e^x (-A \sin x + B \cos x)$$

$$\Rightarrow \frac{d^2y}{dx^2} - \frac{dy}{dx} + y = \frac{dy}{dx} - y$$

$$\Rightarrow \frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + 2y = 0$$

(ix) The solution of the differential equation $\frac{dy}{dx} = \frac{x+2y}{x}$ is $x+y = kx^2$

Ans: True, given that $\frac{dy}{dx} = \frac{x+2y}{x}$

$$\frac{dy}{dx} = 1 + \frac{2}{x}y$$

$$\Rightarrow \frac{dy}{dx} - \frac{2}{x}y = 1$$

$$\text{IF} = e^{\int -\frac{2}{x} dx} = e^{-2 \log x} = x^{-2}$$

Differential solution is $y \times x^{-2} = \int x^{-2} \times 1 dx + k$

$$\frac{y}{x^2} = \frac{x^{-2+1}}{-2+1} + k$$

$$\Rightarrow \frac{y}{x^2} = \frac{-1}{x} + k$$

$$\Rightarrow y = -x + kx^2$$

$$\Rightarrow x+y = kx^2$$

(x) Solution of $\frac{x dy}{dx} = y + x \tan \frac{y}{x}$ is $\sin\left(\frac{y}{x}\right) = cx$

Ans: True, given $\frac{x dy}{dx} = y + x \tan \frac{y}{x}$

$$\frac{dy}{dx} = \frac{y}{x} + \tan\left(\frac{y}{x}\right)$$

Put $\frac{y}{x} = v$ i.e. $y = vx$

$$\frac{dx}{x} = v + \frac{xdv}{dx}$$

Substitute values we get

$$\frac{xdv}{dx} + v = v + \tan v$$

$$\Rightarrow \frac{dx}{x} = \frac{dv}{\tan v}$$

Integrate both sides

$$\int \frac{1}{x} dx = \int \frac{1}{\tan v} dx$$

$$\Rightarrow \log(x) = \log(\sin v) + \log C'$$

$$\Rightarrow \log\left(\frac{x}{\sin v}\right) = \log C'$$

$$\Rightarrow \frac{x}{\sin v} = C'$$

$$\Rightarrow \sin v = Cx \quad \left[\text{where } C = \frac{1}{C'} \right]$$

$$\Rightarrow \sin \frac{y}{x} = Cx$$

(xi) The differential equation of all non-horizontal lines in a plane is

$$\frac{d^2x}{dy^2} = 0 \times \frac{d^2y}{dx^2} = 0$$

Ans: True

Let non-horizontal line in a plane is $y = mx + c$

Differentiate both sides, we get:

$$\frac{dy}{dx} = m$$

Now, again differentiate both sides, we get:

$$\frac{d^2y}{dx^2} = 0$$